

# KIT SAFETY DATA SHEET



Kit Product Name Monolisa Anti-HBs PLUS, 192 Tests  
Kit Catalog Number(s) 72566  
Revision date 22-Aug-2024

## Kit Contents

| Catalog Number(s)          | Product Name                                          |
|----------------------------|-------------------------------------------------------|
| 7361E, 7360T, 7361J        | R2 - 20 x Conc. Washing Solution, 235 ml              |
| 7360G, 5180S, 7435S        | R8 - Substrat Buffer, 60 mL                           |
| 7360J, 5180U, 7361H, 7337Z | R10 - Stopping Solution, 28 mL                        |
| 7436L, 7436H               | R9 - Chromogen: TMB Solution (11X), 5 mL              |
| 7441A                      | R1 - Microplate, 12 x 8 wells                         |
| 7441C                      | C0 - Negative Control, 2.2 ml                         |
| 7441E                      | C1 - 10 mIU/ml Calibrator, 3 ml                       |
| 7441D                      | C2 - 100 mIU/ml Calibrator / Positive Control, 2.2 ml |
| 7441V                      | C3 - 400 mIU/ml Calibrator, 2.2 ml                    |
| 7441F                      | C4 - 1000 mIU/ml Calibrator, 2.2 ml                   |
| 7441B                      | R6 - Specimen Diluent, 27 ml                          |
| 7441G                      | R7a - Conjugate (11X), 2.5 ml                         |
| 7441H                      | R7b - Conjugate Diluent, 25 ml                        |



# SAFETY DATA SHEET

This safety data sheet was created pursuant to the requirements of:  
Canada WHMIS 2015 which includes the amended Hazardous Products Act (HPA) and the Hazardous Products Regulation (HPR)

Revision date 21-Aug-2024

Revision Number 1.2

## 1. Identification

### Product identifier

**Product Name** R2 - 20 x Conc. Washing Solution, 235 ml

### Other means of identification

**Catalog Number(s)** 7361E, 7360T, 7361J

### Recommended use of the chemical and restrictions on use

**Recommended use** Restricted to professional users  
In vitro diagnostic

**Restrictions on use** No information available

### Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer Address

Bio-Rad  
3 boulevard Raymond Poincaré  
92430 Marnes-la-Coquette  
France  
e-mail: fds-msds.fr@bio-rad.com

#### Legal Entity / Contact Address

Bio-Rad Laboratories (Canada) Ltd.  
2403 Guenette  
Montreal, Quebec H4R 2E9  
Canada

**Technical Service** 1-800-361-1808  
CSD\_Techsupport@bio-rad.com

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC Canada: 1 (800) 424-9300

## 2. Hazard(s) identification

### Classification

Not classified

### Label elements

### **Hazard statements**

Not classified.

### Other information

No information available.

### 3. Composition/information on ingredients

#### Substance

Not applicable.

#### Mixture

The product contains no substances which at their given concentration, are considered to be hazardous to health

| Chemical name                                                                    | CAS No.    | Weight-%     | Hazardous Material Information Review Act registry number (HMIRA registry #) | Date HMIRA filed and date exemption granted (if applicable) |
|----------------------------------------------------------------------------------|------------|--------------|------------------------------------------------------------------------------|-------------------------------------------------------------|
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with 2-methyl-3(2H)-isothiazolone | 55965-84-9 | 0.001 - 0.01 | -                                                                            |                                                             |

### 4. First-aid measures

#### Description of first aid measures

|                |                                                                                                                      |
|----------------|----------------------------------------------------------------------------------------------------------------------|
| General advice | No hazards which require special first aid measures.                                                                 |
| Inhalation     | Remove to fresh air.                                                                                                 |
| Eye contact    | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids. Consult a physician. |
| Skin contact   | Wash skin with soap and water.                                                                                       |
| Ingestion      | Rinse mouth thoroughly with water.                                                                                   |

#### Most important symptoms and effects, both acute and delayed

|          |                           |
|----------|---------------------------|
| Symptoms | No information available. |
|----------|---------------------------|

#### Indication of any immediate medical attention and special treatment needed

|                    |                        |
|--------------------|------------------------|
| Note to physicians | Treat symptomatically. |
|--------------------|------------------------|

### 5. Fire-fighting measures

|                                                |                                                                                                                                    |
|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Suitable Extinguishing Media                   | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.                            |
| Unsuitable extinguishing media                 | No information available.                                                                                                          |
| Specific hazards arising from the chemical     | None known.                                                                                                                        |
| Explosion data                                 |                                                                                                                                    |
| Sensitivity to mechanical impact               | None.                                                                                                                              |
| Sensitivity to static discharge                | None.                                                                                                                              |
| Special protective equipment for fire-fighters | Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment. |

## 6. Accidental release measures

### Personal precautions, protective equipment and emergency procedures

**Personal precautions** See section 8 for more information.

### Methods and material for containment and cleaning up

**Methods for containment** Prevent further leakage or spillage if safe to do so.

**Methods for cleaning up** Pick up and transfer to properly labeled containers.

## 7. Handling and storage

### Precautions for safe handling

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice.

### Conditions for safe storage, including any incompatibilities

**Storage Conditions** Store according to product and label instructions.

## 8. Exposure controls/personal protection

### Control parameters

**Exposure Limits** This product, as supplied, does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.

### Appropriate engineering controls

**Engineering controls** Showers  
Eyewash stations  
Ventilation systems.

### Individual protection measures, such as personal protective equipment

**Eye/face protection** Wear safety glasses with side shields (or goggles).

**Hand protection** Wear suitable gloves.

**Skin and body protection** Wear suitable protective clothing.

**Respiratory protection** No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

**General hygiene considerations** Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

### Information on basic physical and chemical properties

**Physical state** Liquid  
**Appearance** Liquid  
**Color** colorless

Odor Odorless  
 Odor threshold No information available

| <u>Property</u>                         | <u>Values</u>     | <u>Remarks • Method</u> |
|-----------------------------------------|-------------------|-------------------------|
| pH                                      |                   |                         |
| Melting point / freezing point          | No data available | None known              |
| Initial boiling point and boiling range | No data available | None known              |
| Flash point                             | No data available | None known              |
| Evaporation rate                        | No data available | None known              |
| Flammability                            | No data available | None known              |
| Flammability Limit in Air               |                   | None known              |
| Upper flammability or explosive limits  | No data available |                         |
| Lower flammability or explosive limits  | No data available |                         |
| Vapor pressure                          | No data available | None known              |
| Relative vapor density                  | No data available | None known              |
| Relative density                        | No data available | None known              |
| Water solubility                        | Miscible in water |                         |
| Solubility in other solvents            | No data available | None known              |
| Partition coefficient                   | No data available | None known              |
| Autoignition temperature                | No data available | None known              |
| Decomposition temperature               |                   | None known              |
| Kinematic viscosity                     | No data available | None known              |
| Dynamic viscosity                       | No data available | None known              |
| <u>Other information</u>                |                   |                         |
| Explosive properties                    | Not applicable.   |                         |
| Oxidizing properties                    | Not applicable.   |                         |
| Softening point                         | Not applicable    |                         |
| Molecular weight                        | Not applicable    |                         |
| VOC content                             | Not applicable    |                         |

## 10. Stability and reactivity

Reactivity No information available.

Chemical stability Stable under normal conditions.

Possibility of hazardous reactions None under normal processing.

Conditions to avoid None known based on information supplied.

Incompatible materials None known based on information supplied.

Hazardous decomposition products None known based on information supplied.

## 11. Toxicological information

### Information on likely routes of exposure

#### Product Information

Inhalation Specific test data for the substance or mixture is not available.

Eye contact Specific test data for the substance or mixture is not available.

Skin contact Specific test data for the substance or mixture is not available.

Ingestion Specific test data for the substance or mixture is not available.

### Symptoms related to the physical, chemical and toxicological characteristics

**Symptoms** No information available.

#### Acute toxicity

#### **Numerical measures of toxicity**

No information available

The following values are calculated based on chapter 3.1 of the GHS document

ATEmix (oral) 11,155.50 mg/kg  
ATEmix (dermal) 206,611.60 mg/kg

#### **Component Information**

| Chemical name                                                                               | Oral LD50          | Dermal LD50              | Inhalation LC50 |
|---------------------------------------------------------------------------------------------|--------------------|--------------------------|-----------------|
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with 2-methyl-3(2H)-isothiazolone 55965-84-9 | = 53 mg/kg ( Rat ) | = 87.12 mg/kg ( Rabbit ) | -               |

#### Delayed and immediate effects as well as chronic effects from short and long-term exposure

**Skin corrosion/irritation** Based on available data, the classification criteria are not met.

**Serious eye damage/eye irritation** Based on available data, the classification criteria are not met.

**Respiratory or skin sensitization** Based on available data, the classification criteria are not met.

**Germ cell mutagenicity** Based on available data, the classification criteria are not met.

**Carcinogenicity** Based on available data, the classification criteria are not met.

**Reproductive toxicity** Based on available data, the classification criteria are not met.

**STOT - single exposure** Based on available data, the classification criteria are not met.

**STOT - repeated exposure** Based on available data, the classification criteria are not met.

**Aspiration hazard** Based on available data, the classification criteria are not met.

## **12. Ecological information**

#### **Ecotoxicity**

**Persistence and degradability** No information available.

**Bioaccumulation** There is no data for this product.

#### **Component Information**

| Chemical name                                                                               | Partition coefficient |
|---------------------------------------------------------------------------------------------|-----------------------|
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with 2-methyl-3(2H)-isothiazolone 55965-84-9 | 0.7                   |

**Other adverse effects** No information available.

## **13. Disposal considerations**

#### Disposal methods

|                                            |                                                                                                                 |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>Waste from residues/unused products</b> | Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation. |
| <b>Contaminated packaging</b>              | Do not reuse empty containers.                                                                                  |

## 14. Transport information

|             |               |
|-------------|---------------|
| <b>TDG</b>  | Not regulated |
| <b>DOT</b>  | Not regulated |
| <b>MEX</b>  | Not regulated |
| <b>IATA</b> | Not regulated |
| <b>IMDG</b> | Not regulated |

## 15. Regulatory information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Regulations

The Montreal Protocol on Substances that Deplete the Ozone Layer Not applicable

The Stockholm Convention on Persistent Organic Pollutants Not applicable

The Rotterdam Convention Not applicable

#### International Inventories

Contact supplier for inventory compliance status

## 16. Other information

|                                   |                         |                                  |                           |                                           |
|-----------------------------------|-------------------------|----------------------------------|---------------------------|-------------------------------------------|
| <b>NFPA</b>                       | <b>Health hazards</b> 0 | <b>Flammability</b> 0            | <b>Instability</b> 0      | <b>Physical and chemical properties -</b> |
| <b>HMIS</b>                       | <b>Health hazards</b> 0 | <b>Flammability</b> 0            | <b>Physical hazards</b> 0 | <b>Personal protection</b> X              |
| <i>Chronic Hazard Star Legend</i> |                         | <i>* = Chronic Health Hazard</i> |                           |                                           |

### Key or legend to abbreviations and acronyms used in the safety data sheet

#### **Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | Sk*  | Skin designation                 |

#### **Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
 U.S. Environmental Protection Agency ChemView Database  
 European Food Safety Authority (EFSA)  
 Environmental Protection Agency  
 Acute Exposure Guideline Level(s) (AEGL(s))  
 U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
 U.S. Environmental Protection Agency High Production Volume Chemicals  
 Food Research Journal  
 Hazardous Substance Database  
 International Uniform Chemical Information Database (IUCLID)  
 Japan GHS Classification

Australia National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
NIOSH (National Institute for Occupational Safety and Health)  
National Library of Medicine's ChemID Plus (NLM CIP)  
National Library of Medicine's PubMed database (NLM PUBMED)  
U.S. National Toxicology Program (NTP)  
New Zealand's Chemical Classification and Information Database (CCID)  
Organization for Economic Co-operation and Development Environment, Health, and Safety Publications  
Organization for Economic Co-operation and Development High Production Volume Chemicals Program  
Organization for Economic Co-operation and Development Screening Information Data Set  
RTECS (Registry of Toxic Effects of Chemical Substances)  
World Health Organization

**Prepared By** Bio-Rad Laboratories, Environmental Health and Safety.

**Revision date** 21-Aug-2024

**Revision Note** Significant changes throughout SDS. Review all sections.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

**End of Safety Data Sheet**





# SAFETY DATA SHEET

This safety data sheet was created pursuant to the requirements of:  
Canada WHMIS 2015 which includes the amended Hazardous Products Act (HPA) and the Hazardous Products Regulation (HPR)

Revision date 09-Aug-2024

Revision Number 2.3

## 1. Identification

### Product identifier

Product Name R8 - Substrat Buffer, 60 mL

### Other means of identification

Catalog Number(s) 7360G, 5180S, 7435S

### Recommended use of the chemical and restrictions on use

Recommended use In vitro diagnostic  
Restricted to professional users

Restrictions on use No information available

### Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer Address

Bio-Rad  
3 boulevard Raymond Poincaré  
92430 Marnes-la-Coquette  
France  
e-mail: fds-msds.fr@bio-rad.com

#### Legal Entity / Contact Address

Bio-Rad Laboratories (Canada) Ltd.  
2403 Guenette  
Montreal, Quebec H4R 2E9  
Canada

Technical Service 1-800-361-1808  
CSD\_Techsupport@bio-rad.com

### Emergency telephone number

24 Hour Emergency Phone Number CHEMTREC Canada:1 (800) 424-9300

## 2. Hazard(s) identification

### Classification

Not classified

Acute toxicity - Inhalation (Dusts/Mists)

Not applicable

### Label elements

#### **Hazard statements**

Not classified.

### Other information

No information available.

### 3. Composition/information on ingredients

#### Substance

Not applicable.

#### Mixture

The product contains no substances which at their given concentration, are considered to be hazardous to health

### 4. First-aid measures

#### Description of first aid measures

|                |                                                                                                                         |
|----------------|-------------------------------------------------------------------------------------------------------------------------|
| General advice | No hazards which require special first aid measures.                                                                    |
| Inhalation     | Remove to fresh air.                                                                                                    |
| Eye contact    | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids.<br>Consult a physician. |
| Skin contact   | Wash skin with soap and water.                                                                                          |
| Ingestion      | Rinse mouth thoroughly with water.                                                                                      |

#### Most important symptoms and effects, both acute and delayed

|          |                           |
|----------|---------------------------|
| Symptoms | No information available. |
|----------|---------------------------|

#### Indication of any immediate medical attention and special treatment needed

|                    |                        |
|--------------------|------------------------|
| Note to physicians | Treat symptomatically. |
|--------------------|------------------------|

### 5. Fire-fighting measures

|                                                |                                                                                                                                       |
|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Suitable Extinguishing Media                   | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.                               |
| Unsuitable extinguishing media                 | No information available.                                                                                                             |
| Specific hazards arising from the chemical     | None known.                                                                                                                           |
| Explosion data                                 |                                                                                                                                       |
| Sensitivity to mechanical impact               | None.                                                                                                                                 |
| Sensitivity to static discharge                | None.                                                                                                                                 |
| Special protective equipment for fire-fighters | Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear.<br>Use personal protection equipment. |

### 6. Accidental release measures

#### Personal precautions, protective equipment and emergency procedures

**Personal precautions** See section 8 for more information.

**Methods and material for containment and cleaning up**

**Methods for containment** Prevent further leakage or spillage if safe to do so.

**Methods for cleaning up** Pick up and transfer to properly labeled containers.

## 7. Handling and storage

**Precautions for safe handling**

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice.

**Conditions for safe storage, including any incompatibilities**

**Storage Conditions** Store according to product and label instructions.

## 8. Exposure controls/personal protection

**Control parameters**

**Exposure Limits** This product, as supplied, does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.

**Appropriate engineering controls**

**Engineering controls** Showers  
Eyewash stations  
Ventilation systems.

**Individual protection measures, such as personal protective equipment**

**Eye/face protection** Wear safety glasses with side shields (or goggles).

**Hand protection** Wear suitable gloves.

**Skin and body protection** Wear suitable protective clothing.

**Respiratory protection** No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

**General hygiene considerations** Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

**Information on basic physical and chemical properties**

|                       |                          |
|-----------------------|--------------------------|
| <b>Physical state</b> | Liquid                   |
| <b>Appearance</b>     | aqueous solution         |
| <b>Color</b>          | colorless                |
| <b>Odor</b>           | Odorless                 |
| <b>Odor threshold</b> | No information available |

| <b><u>Property</u></b> | <b><u>Values</u></b> | <b><u>Remarks • Method</u></b> |
|------------------------|----------------------|--------------------------------|
| pH                     | 4                    |                                |

|                                         |                             |            |
|-----------------------------------------|-----------------------------|------------|
| Melting point / freezing point          | No data available           | None known |
| Initial boiling point and boiling range | No data available           | None known |
| Flash point                             | No data available           | None known |
| Evaporation rate                        | No data available           | None known |
| Flammability                            | No data available           | None known |
| Flammability Limit in Air               |                             | None known |
| Upper flammability or explosive limits  | No data available           |            |
| Lower flammability or explosive limits  | No data available           |            |
| Vapor pressure                          | No data available           | None known |
| Relative vapor density                  | No data available           | None known |
| Relative density                        | No data available           | None known |
| Water solubility                        | Miscible in water           |            |
| Solubility in other solvents            | No data available           | None known |
| Partition coefficient                   | No data available           | None known |
| Autoignition temperature                | No data available / 1850 °F |            |
| Decomposition temperature               |                             | None known |
| Kinematic viscosity                     | No data available           | None known |
| Dynamic viscosity                       | No data available           | None known |
| <b><u>Other information</u></b>         |                             |            |
| Explosive properties                    | Not applicable.             |            |
| Oxidizing properties                    | Not applicable.             |            |
| Softening point                         | Not applicable              |            |
| Molecular weight                        | Not applicable              |            |
| VOC content                             | Not applicable              |            |

## 10. Stability and reactivity

|                                    |                                           |
|------------------------------------|-------------------------------------------|
| Reactivity                         | No information available.                 |
| Chemical stability                 | Stable under normal conditions.           |
| Possibility of hazardous reactions | None under normal processing.             |
| Conditions to avoid                | None known based on information supplied. |
| Incompatible materials             | None known based on information supplied. |
| Hazardous decomposition products   | None known based on information supplied. |

## 11. Toxicological information

### Information on likely routes of exposure

#### Product Information

|              |                                                                   |
|--------------|-------------------------------------------------------------------|
| Inhalation   | Specific test data for the substance or mixture is not available. |
| Eye contact  | Specific test data for the substance or mixture is not available. |
| Skin contact | Specific test data for the substance or mixture is not available. |
| Ingestion    | Specific test data for the substance or mixture is not available. |

### Symptoms related to the physical, chemical and toxicological characteristics

|          |                           |
|----------|---------------------------|
| Symptoms | No information available. |
|----------|---------------------------|

### Acute toxicity

**Numerical measures of toxicity**

The following values are calculated based on chapter 3.1 of the GHS document

|                               |                  |
|-------------------------------|------------------|
| ATEmix (oral)                 | 82,278.50 mg/kg  |
| ATEmix (dermal)               | 113,895.20 mg/kg |
| ATEmix (inhalation-dust/mist) | 551.50 mg/l      |

**Component Information****Delayed and immediate effects as well as chronic effects from short and long-term exposure**

|                                   |                                                                   |
|-----------------------------------|-------------------------------------------------------------------|
| Skin corrosion/irritation         | Based on available data, the classification criteria are not met. |
| Serious eye damage/eye irritation | Based on available data, the classification criteria are not met. |
| Respiratory or skin sensitization | Based on available data, the classification criteria are not met. |
| Germ cell mutagenicity            | Based on available data, the classification criteria are not met. |
| Carcinogenicity                   | Based on available data, the classification criteria are not met. |
| Reproductive toxicity             | Based on available data, the classification criteria are not met. |
| STOT - single exposure            | Based on available data, the classification criteria are not met. |
| STOT - repeated exposure          | Based on available data, the classification criteria are not met. |
| Aspiration hazard                 | Based on available data, the classification criteria are not met. |

**12. Ecological information****Ecotoxicity**

|                               |                                    |
|-------------------------------|------------------------------------|
| Persistence and degradability | No information available.          |
| Bioaccumulation               | There is no data for this product. |
| Other adverse effects         | No information available.          |

**13. Disposal considerations****Disposal methods**

|                                     |                                                                                                                 |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Waste from residues/unused products | Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation. |
| Contaminated packaging              | Do not reuse empty containers.                                                                                  |

**14. Transport information**

|            |               |
|------------|---------------|
| <u>TDG</u> | Not regulated |
| <u>DOT</u> | Not regulated |

MEX Not regulated

IATA Not regulated

IMDG Not regulated

## 15. Regulatory information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Regulations

The Montreal Protocol on Substances that Deplete the Ozone Layer Not applicable

The Stockholm Convention on Persistent Organic Pollutants Not applicable

The Rotterdam Convention Not applicable

#### International Inventories

Contact supplier for inventory compliance status

## 16. Other information

|             |                  |                |                    |                                    |
|-------------|------------------|----------------|--------------------|------------------------------------|
| <u>NFPA</u> | Health hazards 0 | Flammability 0 | Instability 0      | Physical and chemical properties - |
| <u>HMIS</u> | Health hazards 0 | Flammability 0 | Physical hazards 0 | Personal protection X              |

### Key or legend to abbreviations and acronyms used in the safety data sheet

#### **Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | Sk*  | Skin designation                 |

#### **Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
 U.S. Environmental Protection Agency ChemView Database  
 European Food Safety Authority (EFSA)  
 Environmental Protection Agency  
 Acute Exposure Guideline Level(s) (AEGl(s))  
 U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
 U.S. Environmental Protection Agency High Production Volume Chemicals  
 Food Research Journal  
 Hazardous Substance Database  
 International Uniform Chemical Information Database (IUCLID)  
 Japan GHS Classification  
 Australia National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
 NIOSH (National Institute for Occupational Safety and Health)  
 National Library of Medicine's ChemID Plus (NLM CIP)  
 National Library of Medicine's PubMed database (NLM PUBMED)  
 U.S. National Toxicology Program (NTP)  
 New Zealand's Chemical Classification and Information Database (CCID)  
 Organization for Economic Co-operation and Development Environment, Health, and Safety Publications  
 Organization for Economic Co-operation and Development High Production Volume Chemicals Program  
 Organization for Economic Co-operation and Development Screening Information Data Set  
 RTECS (Registry of Toxic Effects of Chemical Substances)  
 World Health Organization

**Prepared By** Bio-Rad Laboratories, Environmental Health and Safety.

**Revision date** 09-Aug-2024

**Revision Note**

Significant changes throughout SDS. Review all sections.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

**End of Safety Data Sheet**



# SAFETY DATA SHEET

This safety data sheet was created pursuant to the requirements of:  
Canada WHMIS 2015 which includes the amended Hazardous Products Act (HPA) and the Hazardous Products Regulation (HPR)

Revision date 09-Aug-2024

Revision Number 2.2

## 1. Identification

### Product identifier

Product Name R10 - Stopping Solution, 28 mL

### Other means of identification

Catalog Number(s) 7360J, 5180U, 7361H, 7337Z

UN/ID no UN2796

### Recommended use of the chemical and restrictions on use

Recommended use In vitro diagnostic  
Restricted to professional users

Restrictions on use No information available

### Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer Address

Bio-Rad  
3 boulevard Raymond Poincaré  
92430 Marnes-la-Coquette  
France  
e-mail: fds-msds.fr@bio-rad.com

#### Legal Entity / Contact Address

Bio-Rad Laboratories (Canada) Ltd.  
2403 Guenette  
Montreal, Quebec H4R 2E9  
Canada

Technical Service 1-800-361-1808  
CSD\_Techsupport@bio-rad.com

### Emergency telephone number

24 Hour Emergency Phone Number CHEMTREC Canada:1 (800) 424-9300

## 2. Hazard(s) identification

### Classification

Skin corrosion/irritation

Category 1

### Label elements

Danger

### Hazard statements

Causes severe skin burns and eye damage



**Precautionary Statements - Prevention**

Do not breathe dust/fume/gas/mist/vapors/spray  
Wash face, hands and any exposed skin thoroughly after handling  
Wear protective gloves/protective clothing/eye protection/face protection

**Precautionary Statements - Response**

Immediately call a POISON CENTER or doctor

**Eyes**

Immediately call a POISON CENTER or doctor  
IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

**Skin**

IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water [or shower]  
Wash contaminated clothing before reuse

**Inhalation**

IF INHALED: Remove person to fresh air and keep comfortable for breathing  
Immediately call a POISON CENTER or doctor

**Ingestion**

IF SWALLOWED: Rinse mouth. Do NOT induce vomiting

**Precautionary Statements - Disposal**

Dispose of contents/container in accordance with local, regional, national, and international regulations as applicable

**Other information**

No information available.

**3. Composition/information on ingredients****Substance**

Not applicable.

**Mixture**

| Chemical name | CAS No.   | Weight-% | Hazardous Material<br>Information Review Act<br>registry number<br>(HMIRA registry #) | Date HMIRA filed and<br>date exemption granted<br>(if applicable) |
|---------------|-----------|----------|---------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| Sulfuric acid | 7664-93-9 | 2.5 - 5  | -                                                                                     |                                                                   |

**4. First-aid measures****Description of first aid measures**

|                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General advice</b>                     | Immediate medical attention is required. Show this safety data sheet to the doctor in attendance.                                                                                                                                                                                                                                                                                                                                      |
| <b>Inhalation</b>                         | Remove to fresh air. If breathing has stopped, give artificial respiration. Get medical attention immediately. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. If breathing is difficult, (trained personnel should) give oxygen. Delayed pulmonary edema may occur. |
| <b>Eye contact</b>                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Keep eye wide open while rinsing. Do not rub affected area. Remove contact lenses, if present and easy to do. Continue rinsing. Get immediate medical advice/attention.                                                                                                                                                                       |
| <b>Skin contact</b>                       | Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Get immediate medical advice/attention.                                                                                                                                                                                                                                                                                          |
| <b>Ingestion</b>                          | Do NOT induce vomiting. Clean mouth with water and drink afterwards plenty of water. Never give anything by mouth to an unconscious person. Get immediate medical advice/attention.                                                                                                                                                                                                                                                    |
| <b>Self-protection of the first aider</b> | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination. Avoid contact with skin, eyes or clothing. Avoid direct contact with skin. Use barrier to give mouth-to-mouth resuscitation. Wear personal protective clothing (see section 8).                                                                                                       |

**Most important symptoms and effects, both acute and delayed**

**Symptoms** Burning sensation.

**Indication of any immediate medical attention and special treatment needed**

**Note to physicians** Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Do not give chemical antidotes. Asphyxia from glottal edema may occur. Marked decrease in blood pressure may occur with moist rales, frothy sputum, and high pulse pressure.

**5. Fire-fighting measures**

|                                                       |                                                                                                                                        |
|-------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>                   | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.                                |
| <b>Unsuitable extinguishing media</b>                 | No information available.                                                                                                              |
| <b>Specific hazards arising from the chemical</b>     | The product causes burns of eyes, skin and mucous membranes. Thermal decomposition can lead to release of irritating gases and vapors. |
| <b>Explosion data</b>                                 |                                                                                                                                        |
| <b>Sensitivity to mechanical impact</b>               | None.                                                                                                                                  |
| <b>Sensitivity to static discharge</b>                | None.                                                                                                                                  |
| <b>Special protective equipment for fire-fighters</b> | Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment.     |

**6. Accidental release measures****Personal precautions, protective equipment and emergency procedures**

**Personal precautions** Attention! Corrosive material. Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak.

**Other information** Refer to protective measures listed in Sections 7 and 8.

**Methods and material for containment and cleaning up**

**Methods for containment** Prevent further leakage or spillage if safe to do so.

**Methods for cleaning up** Pick up and transfer to properly labeled containers.

## 7. Handling and storage

**Precautions for safe handling**

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice. Avoid contact with skin, eyes or clothing. In case of insufficient ventilation, wear suitable respiratory equipment. Handle product only in closed system or provide appropriate exhaust ventilation. Do not eat, drink or smoke when using this product. Take off contaminated clothing and wash before reuse.

**Conditions for safe storage, including any incompatibilities**

**Storage Conditions** Keep containers tightly closed in a dry, cool and well-ventilated place. Protect from moisture. Store locked up. Keep out of the reach of children. Store away from other materials. Store according to product and label instructions.

## 8. Exposure controls/personal protection

**Control parameters**

**Exposure Limits**

| Chemical name              | Alberta                                               | British Columbia           | Ontario                    | Quebec                                                |
|----------------------------|-------------------------------------------------------|----------------------------|----------------------------|-------------------------------------------------------|
| Sulfuric acid<br>7664-93-9 | TWA: 1 mg/m <sup>3</sup><br>STEL: 3 mg/m <sup>3</sup> | TWA: 0.2 mg/m <sup>3</sup> | TWA: 0.2 mg/m <sup>3</sup> | TWA: 1 mg/m <sup>3</sup><br>STEL: 3 mg/m <sup>3</sup> |

**Appropriate engineering controls**

**Engineering controls** Showers  
Eyewash stations  
Ventilation systems.

**Individual protection measures, such as personal protective equipment**

**Eye/face protection** Tight sealing safety goggles. Face protection shield.

**Hand protection** Wear suitable gloves. Impervious gloves.

**Skin and body protection** Wear suitable protective clothing. Long sleeved clothing. Chemical resistant apron.

**Respiratory protection** No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

**General hygiene considerations** Avoid contact with skin, eyes or clothing. Wear suitable gloves and eye/face protection. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Contaminated work clothing should not be allowed out of the workplace. Regular cleaning of equipment, work area and clothing is recommended. Wash hands before breaks and immediately after handling the product.

## 9. Physical and chemical properties

### Information on basic physical and chemical properties

|                |                          |
|----------------|--------------------------|
| Physical state | Liquid                   |
| Appearance     | aqueous solution         |
| Color          | colorless                |
| Odor           | Low                      |
| Odor threshold | No information available |

| Property                                | Values            | Remarks • Method |
|-----------------------------------------|-------------------|------------------|
| pH                                      | < 2               |                  |
| Melting point / freezing point          | No data available | None known       |
| Initial boiling point and boiling range | No data available | None known       |
| Flash point                             | No data available | None known       |
| Evaporation rate                        | No data available | None known       |
| Flammability                            | No data available | None known       |
| Flammability Limit in Air               |                   | None known       |
| Upper flammability or explosive limits  | No data available |                  |
| Lower flammability or explosive limits  | No data available |                  |
| Vapor pressure                          | No data available | None known       |
| Relative vapor density                  | No data available | None known       |
| Relative density                        | No data available | None known       |
| Water solubility                        | Miscible in water |                  |
| Solubility in other solvents            | No data available | None known       |
| Partition coefficient                   | No data available | None known       |
| Autoignition temperature                | No data available | None known       |
| Decomposition temperature               |                   | None known       |
| Kinematic viscosity                     | No data available | None known       |
| Dynamic viscosity                       | No data available | None known       |
| <b>Other information</b>                |                   |                  |
| Explosive properties                    | Not applicable.   |                  |
| Oxidizing properties                    | Not applicable.   |                  |
| Softening point                         | Not applicable    |                  |
| Molecular weight                        | Not applicable    |                  |
| VOC content                             | Not applicable    |                  |

## 10. Stability and reactivity

|                                    |                                                     |
|------------------------------------|-----------------------------------------------------|
| Reactivity                         | No information available.                           |
| Chemical stability                 | Stable under normal conditions.                     |
| Possibility of hazardous reactions | None under normal processing.                       |
| Conditions to avoid                | Exposure to air or moisture over prolonged periods. |
| Incompatible materials             | Acids. Bases. Oxidizing agent.                      |
| Hazardous decomposition products   | None known based on information supplied.           |

## 11. Toxicological information

### Information on likely routes of exposure

#### Product Information

|            |                                                                                          |
|------------|------------------------------------------------------------------------------------------|
| Inhalation | Specific test data for the substance or mixture is not available Corrosive by inhalation |
|------------|------------------------------------------------------------------------------------------|

(based on components). Inhalation of corrosive fumes/gases may cause coughing, choking, headache, dizziness, and weakness for several hours. Pulmonary edema may occur with tightness in the chest, shortness of breath, bluish skin, decreased blood pressure, and increased heart rate. Inhaled corrosive substances can lead to a toxic edema of the lungs. Pulmonary edema can be fatal.

**Eye contact** Specific test data for the substance or mixture is not available Causes serious eye damage (based on components). Corrosive to the eyes and may cause severe damage including blindness.

**Skin contact** Specific test data for the substance or mixture is not available Corrosive (based on components). Causes burns.

**Ingestion** Specific test data for the substance or mixture is not available Causes burns (based on components). Ingestion causes burns of the upper digestive and respiratory tracts. May cause severe burning pain in the mouth and stomach with vomiting and diarrhea of dark blood. Blood pressure may decrease. Brownish or yellowish stains may be seen around the mouth. Swelling of the throat may cause shortness of breath and choking. May cause lung damage if swallowed. May be fatal if swallowed and enters airways.

#### Symptoms related to the physical, chemical and toxicological characteristics

**Symptoms** Redness. Burning. May cause blindness. Coughing and/ or wheezing.

#### Acute toxicity

#### **Numerical measures of toxicity**

No information available

#### **Component Information**

| Chemical name              | Oral LD50            | Dermal LD50 | Inhalation LC50          |
|----------------------------|----------------------|-------------|--------------------------|
| Sulfuric acid<br>7664-93-9 | = 2140 mg/kg ( Rat ) | -           | = 0.375 mg/L ( Rat ) 4 h |

#### Delayed and immediate effects as well as chronic effects from short and long-term exposure

**Skin corrosion/irritation** Classification based on data available for ingredients. Causes burns.

**Serious eye damage/eye irritation** Classification based on data available for ingredients. Risk of serious damage to eyes.

**Respiratory or skin sensitization** Based on available data, the classification criteria are not met.

**Germ cell mutagenicity** Based on available data, the classification criteria are not met.

**Carcinogenicity** Based on available data, the classification criteria are not met.

The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Chemical name              | ACGIH | IARC    | NTP   | OSHA |
|----------------------------|-------|---------|-------|------|
| Sulfuric acid<br>7664-93-9 | A2    | Group 1 | Known | X    |

#### **Legend**

**ACGIH (American Conference of Governmental Industrial Hygienists)**

A2 - Suspected Human Carcinogen

**IARC (International Agency for Research on Cancer)**

Group 1 - Carcinogenic to Humans

**NTP (National Toxicology Program)**

Known - Known Carcinogen

**OSHA (Occupational Safety and Health Administration of the US Department of Labor)**

X - Present

**Reproductive toxicity** Based on available data, the classification criteria are not met.

|                          |                                                                   |
|--------------------------|-------------------------------------------------------------------|
| STOT - single exposure   | Based on available data, the classification criteria are not met. |
| STOT - repeated exposure | Based on available data, the classification criteria are not met. |
| Target organ effects     | Respiratory system, Eyes, Skin, Teeth.                            |
| Aspiration hazard        | Based on available data, the classification criteria are not met. |

## 12. Ecological information

### Ecotoxicity

| Chemical name              | Algae/aquatic plants | Fish                                    | Toxicity to microorganisms | Crustacea |
|----------------------------|----------------------|-----------------------------------------|----------------------------|-----------|
| Sulfuric acid<br>7664-93-9 | -                    | LC50: >500mg/L (96h, Brachydanio rerio) | -                          | -         |

|                               |                           |
|-------------------------------|---------------------------|
| Persistence and degradability | No information available. |
| Bioaccumulation               | No information available. |
| Other adverse effects         | No information available. |

## 13. Disposal considerations

### Disposal methods

|                                     |                                                                                                                 |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Waste from residues/unused products | Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation. |
| Contaminated packaging              | Do not reuse empty containers.                                                                                  |

## 14. Transport information

### TDG

|                            |                                       |
|----------------------------|---------------------------------------|
| UN/ID no                   | UN2796                                |
| UN proper shipping name    | SULFURIC ACID SOLUTION                |
| Transport hazard class(es) | 8                                     |
| Packing group              | II                                    |
| Description                | UN2796, SULFURIC ACID SOLUTION, 8, II |

### DOT

|                                 |                                             |
|---------------------------------|---------------------------------------------|
| UN/ID no                        | UN2796                                      |
| Extended proper shipping name   | SULFURIC ACID SOLUTION                      |
| Transport hazard class(es)      | 8                                           |
| Packing group                   | II                                          |
| Reportable Quantity (RQ)        | (Sulfuric acid: RQ (kg)= 454.00)            |
| Special Provisions              | 386, A3, A7, B2, B15, IB2, N6, N34, T8, TP2 |
| Description                     | UN2796, SULFURIC ACID SOLUTION, 8, II       |
| Emergency Response Guide Number | 157                                         |

### MEX

|                            |                         |
|----------------------------|-------------------------|
| UN/ID no                   | UN2796                  |
| UN proper shipping name    | SULPHURIC ACID SOLUTION |
| Transport hazard class(es) | 8                       |

|                      |                                        |
|----------------------|----------------------------------------|
| <b>Packing group</b> | II                                     |
| <b>Description</b>   | UN2796, SULPHURIC ACID SOLUTION, 8, II |

**IATA**

|                                   |                                        |
|-----------------------------------|----------------------------------------|
| <b>UN number or ID number</b>     | UN2796                                 |
| <b>UN proper shipping name</b>    | Sulphuric acid solution                |
| <b>Transport hazard class(es)</b> | 8                                      |
| <b>Packing group</b>              | II                                     |
| <b>ERG Code</b>                   | 8L                                     |
| <b>Description</b>                | UN2796, Sulphuric acid solution, 8, II |

**IMDG**

|                                   |                                        |
|-----------------------------------|----------------------------------------|
| <b>UN number or ID number</b>     | UN2796                                 |
| <b>UN proper shipping name</b>    | SULPHURIC ACID SOLUTION                |
| <b>Transport hazard class(es)</b> | 8                                      |
| <b>Packing group</b>              | II                                     |
| <b>EmS-No.</b>                    | F-A, S-B                               |
| <b>Marine pollutant</b>           | NP                                     |
| <b>Description</b>                | UN2796, SULPHURIC ACID SOLUTION, 8, II |

**15. Regulatory information****Safety, health and environmental regulations/legislation specific for the substance or mixture****International Regulations**

The Montreal Protocol on Substances that Deplete the Ozone Layer Not applicable

The Stockholm Convention on Persistent Organic Pollutants Not applicable

The Rotterdam Convention Not applicable

**International Inventories**

Contact supplier for inventory compliance status

**16. Other information**

|                                   |                         |                                  |                           |                                           |
|-----------------------------------|-------------------------|----------------------------------|---------------------------|-------------------------------------------|
| <b>NFPA</b>                       | <b>Health hazards</b> 3 | <b>Flammability</b> 0            | <b>Instability</b> 0      | <b>Physical and chemical properties</b> - |
| <b>HMIS</b>                       | <b>Health hazards</b> 3 | <b>Flammability</b> 0            | <b>Physical hazards</b> 0 | <b>Personal protection</b> X              |
| <i>Chronic Hazard Star Legend</i> |                         | <i>* = Chronic Health Hazard</i> |                           |                                           |

**Key or legend to abbreviations and acronyms used in the safety data sheet****Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | Sk*  | Skin designation                 |

**Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
 U.S. Environmental Protection Agency ChemView Database  
 European Food Safety Authority (EFSA)  
 Environmental Protection Agency  
 Acute Exposure Guideline Level(s) (AEGl(s))  
 U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
 U.S. Environmental Protection Agency High Production Volume Chemicals  
 Food Research Journal  
 Hazardous Substance Database  
 International Uniform Chemical Information Database (IUCLID)  
 Japan GHS Classification  
 Australia National Industrial Chemicals Notification and Assessment Scheme (NICNAS)

NIOSH (National Institute for Occupational Safety and Health)  
National Library of Medicine's ChemID Plus (NLM CIP)  
National Library of Medicine's PubMed database (NLM PUBMED)  
U.S. National Toxicology Program (NTP)  
New Zealand's Chemical Classification and Information Database (CCID)  
Organization for Economic Co-operation and Development Environment, Health, and Safety Publications  
Organization for Economic Co-operation and Development High Production Volume Chemicals Program  
Organization for Economic Co-operation and Development Screening Information Data Set  
RTECS (Registry of Toxic Effects of Chemical Substances)  
World Health Organization

**Prepared By** Bio-Rad Laboratories, Environmental Health and Safety.

**Revision date** 09-Aug-2024

**Revision Note** Significant changes throughout SDS. Review all sections.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

**End of Safety Data Sheet**





# SAFETY DATA SHEET

This safety data sheet was created pursuant to the requirements of:  
Canada WHMIS 2015 which includes the amended Hazardous Products Act (HPA) and the Hazardous Products Regulation (HPR)

Revision date 06-Aug-2024

Revision Number 1.3

## 1. Identification

### Product identifier

Product Name R9 - Chromogen: TMB Solution (11X), 5 mL

### Other means of identification

Catalog Number(s) 7436L, 7436H

### Recommended use of the chemical and restrictions on use

Recommended use Restricted to professional users  
In vitro diagnostic

Restrictions on use No information available

### Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer Address

Bio-Rad  
3 boulevard Raymond Poincaré  
92430 Marnes-la-Coquette  
France  
e-mail: fds-msds.fr@bio-rad.com

#### Legal Entity / Contact Address

Bio-Rad Laboratories (Canada) Ltd.  
2403 Guenette  
Montreal, Quebec H4R 2E9  
Canada

#### Technical Service

1-800-361-1808  
CSD\_Techsupport@bio-rad.com

### Emergency telephone number

24 Hour Emergency Phone Number CHEMTREC Canada:1 (800) 424-9300

## 2. Hazard(s) identification

### Classification

Skin corrosion/irritation

Category 1

### Label elements

#### **Danger**

#### **Hazard statements**

Causes severe skin burns and eye damage



**Precautionary Statements - Prevention**

Do not breathe dust/fume/gas/mist/vapors/spray  
Wash face, hands and any exposed skin thoroughly after handling  
Wear protective gloves/protective clothing/eye protection/face protection

**Precautionary Statements - Response**

Immediately call a POISON CENTER or doctor

**Eyes**

Immediately call a POISON CENTER or doctor  
IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

**Skin**

IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water [or shower]  
Wash contaminated clothing before reuse

**Inhalation**

IF INHALED: Remove person to fresh air and keep comfortable for breathing  
Immediately call a POISON CENTER or doctor

**Ingestion**

IF SWALLOWED: Rinse mouth. Do NOT induce vomiting

**Precautionary Statements - Disposal**

Dispose of contents/container in accordance with local, regional, national, and international regulations as applicable

**Other information**

No information available.

**3. Composition/information on ingredients****Substance**

Not applicable.

**Mixture**

The product contains no substances which at their given concentration, are considered to be hazardous to health

| Chemical name     | CAS No.   | Weight-%     | Hazardous Material<br>Information Review Act<br>registry number<br>(HMIRA registry #) | Date HMIRA filed and<br>date exemption granted<br>(if applicable) |
|-------------------|-----------|--------------|---------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| Hydrochloric acid | 7647-01-0 | 0.3 - 0.99   | -                                                                                     |                                                                   |
| Ethyl alcohol     | 64-17-5   | 0.01 - 0.099 | -                                                                                     |                                                                   |

**4. First-aid measures****Description of first aid measures****General advice**

Immediate medical attention is required. Show this safety data sheet to the doctor in attendance.

**Inhalation**

Remove to fresh air. If breathing has stopped, give artificial respiration. Get medical

attention immediately. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. If breathing is difficult, (trained personnel should) give oxygen. Delayed pulmonary edema may occur.

|                                           |                                                                                                                                                                                                                                                                                                                                  |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye contact</b>                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Keep eye wide open while rinsing. Do not rub affected area. Remove contact lenses, if present and easy to do. Continue rinsing. Get immediate medical advice/attention.                                                                 |
| <b>Skin contact</b>                       | Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Get immediate medical advice/attention.                                                                                                                                                                                    |
| <b>Ingestion</b>                          | Do NOT induce vomiting. Clean mouth with water and drink afterwards plenty of water. Never give anything by mouth to an unconscious person. Get immediate medical advice/attention.                                                                                                                                              |
| <b>Self-protection of the first aider</b> | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination. Avoid contact with skin, eyes or clothing. Avoid direct contact with skin. Use barrier to give mouth-to-mouth resuscitation. Wear personal protective clothing (see section 8). |

#### **Most important symptoms and effects, both acute and delayed**

|                 |                    |
|-----------------|--------------------|
| <b>Symptoms</b> | Burning sensation. |
|-----------------|--------------------|

#### **Indication of any immediate medical attention and special treatment needed**

|                           |                                                                                                                                                                                                                                                                                                                                        |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Note to physicians</b> | Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Do not give chemical antidotes. Asphyxia from glottal edema may occur. Marked decrease in blood pressure may occur with moist rales, frothy sputum, and high pulse pressure. |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### **5. Fire-fighting measures**

|                                                       |                                                                                                                                        |
|-------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>                   | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.                                |
| <b>Unsuitable extinguishing media</b>                 | No information available.                                                                                                              |
| <b>Specific hazards arising from the chemical</b>     | The product causes burns of eyes, skin and mucous membranes. Thermal decomposition can lead to release of irritating gases and vapors. |
| <b>Explosion data</b>                                 |                                                                                                                                        |
| <b>Sensitivity to mechanical impact</b>               | None.                                                                                                                                  |
| <b>Sensitivity to static discharge</b>                | None.                                                                                                                                  |
| <b>Special protective equipment for fire-fighters</b> | Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment.     |

### **6. Accidental release measures**

#### **Personal precautions, protective equipment and emergency procedures**

|                             |                                                                                                                                                                                                                                         |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal precautions</b> | Attention! Corrosive material. Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. |
| <b>Other information</b>    | Refer to protective measures listed in Sections 7 and 8.                                                                                                                                                                                |

#### **Methods and material for containment and cleaning up**

**Methods for containment** Prevent further leakage or spillage if safe to do so.

**Methods for cleaning up** Pick up and transfer to properly labeled containers.

## 7. Handling and storage

### Precautions for safe handling

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice. Avoid contact with skin, eyes or clothing. In case of insufficient ventilation, wear suitable respiratory equipment. Handle product only in closed system or provide appropriate exhaust ventilation. Do not eat, drink or smoke when using this product. Take off contaminated clothing and wash before reuse.

### Conditions for safe storage, including any incompatibilities

**Storage Conditions** Keep containers tightly closed in a dry, cool and well-ventilated place. Protect from moisture. Store locked up. Keep out of the reach of children. Store away from other materials. Store according to product and label instructions.

## 8. Exposure controls/personal protection

### Control parameters

#### Exposure Limits

| Chemical name                  | Alberta                                        | British Columbia | Ontario        | Quebec         |
|--------------------------------|------------------------------------------------|------------------|----------------|----------------|
| Hydrochloric acid<br>7647-01-0 | Ceiling: 2 ppm<br>Ceiling: 3 mg/m <sup>3</sup> | Ceiling: 2 ppm   | CEV: 2 ppm     | Ceiling: 2 ppm |
| Ethyl alcohol<br>64-17-5       | TWA: 1000 ppm<br>TWA: 1880 mg/m <sup>3</sup>   | STEL: 1000 ppm   | STEL: 1000 ppm | STEL: 1000 ppm |

### Appropriate engineering controls

**Engineering controls** Showers  
Eyewash stations  
Ventilation systems.

### Individual protection measures, such as personal protective equipment

**Eye/face protection** Tight sealing safety goggles. Face protection shield.

**Hand protection** Wear suitable gloves. Impervious gloves.

**Skin and body protection** Wear suitable protective clothing. Long sleeved clothing. Chemical resistant apron.

**Respiratory protection** No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

**General hygiene considerations** Avoid contact with skin, eyes or clothing. Wear suitable gloves and eye/face protection. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Contaminated work clothing should not be allowed out of the workplace. Regular cleaning of equipment, work area and clothing is recommended. Wash hands before breaks and immediately after handling the product.

## 9. Physical and chemical properties

**Information on basic physical and chemical properties**

|                |                          |
|----------------|--------------------------|
| Physical state | Liquid                   |
| Appearance     | Liquid                   |
| Color          | pink                     |
| Odor           | Low                      |
| Odor threshold | No information available |

| <b>Property</b>                         | <b>Values</b>     | <b>Remarks • Method</b> |
|-----------------------------------------|-------------------|-------------------------|
| pH                                      |                   |                         |
| Melting point / freezing point          | No data available | None known              |
| Initial boiling point and boiling range | No data available | None known              |
| Flash point                             | No data available | None known              |
| Evaporation rate                        | No data available | None known              |
| Flammability                            | No data available | None known              |
| Flammability Limit in Air               |                   | None known              |
| Upper flammability or explosive limits  | No data available |                         |
| Lower flammability or explosive limits  | No data available |                         |
| Vapor pressure                          | No data available | None known              |
| Relative vapor density                  | No data available | None known              |
| Relative density                        | No data available | None known              |
| Water solubility                        | Miscible in water |                         |
| Solubility in other solvents            | No data available | None known              |
| Partition coefficient                   | No data available | None known              |
| Autoignition temperature                | 363 °C / 685.4 °F |                         |
| Decomposition temperature               |                   | None known              |
| Kinematic viscosity                     | No data available | None known              |
| Dynamic viscosity                       | No data available | None known              |
| <b>Other information</b>                |                   |                         |
| Explosive properties                    | Not applicable.   |                         |
| Oxidizing properties                    | Not applicable.   |                         |
| Softening point                         | Not applicable    |                         |
| Molecular weight                        | Not applicable    |                         |
| VOC content                             | Not applicable    |                         |

**10. Stability and reactivity**

|                                    |                                                     |
|------------------------------------|-----------------------------------------------------|
| Reactivity                         | No information available.                           |
| Chemical stability                 | Stable under normal conditions.                     |
| Possibility of hazardous reactions | None under normal processing.                       |
| Conditions to avoid                | Exposure to air or moisture over prolonged periods. |
| Incompatible materials             | Acids. Bases. Oxidizing agent.                      |
| Hazardous decomposition products   | None known based on information supplied.           |

**11. Toxicological information****Information on likely routes of exposure****Product Information**

|                   |                                                                                                                                                                                                                                                                                                                                                              |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b> | Specific test data for the substance or mixture is not available Corrosive by inhalation (based on components). Inhalation of corrosive fumes/gases may cause coughing, choking, headache, dizziness, and weakness for several hours. Pulmonary edema may occur with tightness in the chest, shortness of breath, bluish skin, decreased blood pressure, and |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

increased heart rate. Inhaled corrosive substances can lead to a toxic edema of the lungs. Pulmonary edema can be fatal.

**Eye contact**

Specific test data for the substance or mixture is not available Causes serious eye damage (based on components). Corrosive to the eyes and may cause severe damage including blindness.

**Skin contact**

Specific test data for the substance or mixture is not available Corrosive (based on components). Causes burns.

**Ingestion**

Specific test data for the substance or mixture is not available Causes burns (based on components). Ingestion causes burns of the upper digestive and respiratory tracts. May cause severe burning pain in the mouth and stomach with vomiting and diarrhea of dark blood. Blood pressure may decrease. Brownish or yellowish stains may be seen around the mouth. Swelling of the throat may cause shortness of breath and choking. May cause lung damage if swallowed. May be fatal if swallowed and enters airways.

**Symptoms related to the physical, chemical and toxicological characteristics****Symptoms**

Redness. Burning. May cause blindness. Coughing and/ or wheezing.

**Acute toxicity****Numerical measures of toxicity**

No information available

**Component Information**

| Chemical name                  | Oral LD50               | Dermal LD50             | Inhalation LC50                                      |
|--------------------------------|-------------------------|-------------------------|------------------------------------------------------|
| Hydrochloric acid<br>7647-01-0 | 238 - 277 mg/kg ( Rat ) | > 5010 mg/kg ( Rabbit ) | = 1.68 mg/L ( Rat ) 1 h                              |
| Ethyl alcohol<br>64-17-5       | = 7060 mg/kg ( Rat )    | -                       | = 116.9 mg/L ( Rat ) 4 h<br>= 133.8 mg/L ( Rat ) 4 h |

**Delayed and immediate effects as well as chronic effects from short and long-term exposure****Skin corrosion/irritation**

Classification based on data available for ingredients. Causes burns.

**Serious eye damage/eye irritation**

Classification based on data available for ingredients. Risk of serious damage to eyes.

**Respiratory or skin sensitization**

Based on available data, the classification criteria are not met.

**Germ cell mutagenicity**

Based on available data, the classification criteria are not met.

**Carcinogenicity**

Based on available data, the classification criteria are not met.

The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Chemical name                  | ACGIH | IARC    | NTP   | OSHA |
|--------------------------------|-------|---------|-------|------|
| Hydrochloric acid<br>7647-01-0 | -     | Group 3 | -     | X    |
| Ethyl alcohol<br>64-17-5       | A3    | Group 1 | Known | X    |

**Legend**

**ACGIH (American Conference of Governmental Industrial Hygienists)**

A3 - Animal Carcinogen

**IARC (International Agency for Research on Cancer)**

Group 1 - Carcinogenic to Humans

Group 3 - Not Classifiable as to Carcinogenicity in Humans

**NTP (National Toxicology Program)**

Known - Known Carcinogen

**OSHA (Occupational Safety and Health Administration of the US Department of Labor)**

X - Present

|                          |                                                                   |
|--------------------------|-------------------------------------------------------------------|
| Reproductive toxicity    | Based on available data, the classification criteria are not met. |
| STOT - single exposure   | Based on available data, the classification criteria are not met. |
| STOT - repeated exposure | Based on available data, the classification criteria are not met. |
| Aspiration hazard        | Based on available data, the classification criteria are not met. |

## 12. Ecological information

### Ecotoxicity

| Chemical name            | Algae/aquatic plants | Fish                                                                                                                                                | Toxicity to microorganisms | Crustacea                                                                        |
|--------------------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|----------------------------------------------------------------------------------|
| Ethyl alcohol<br>64-17-5 | -                    | LC50: 12.0 - 16.0mL/L (96h, Oncorhynchus mykiss)<br>LC50: >100mg/L (96h, Pimephales promelas)<br>LC50: 13400 - 15100mg/L (96h, Pimephales promelas) | -                          | LC50: 9268 - 14221mg/L (48h, Daphnia magna)<br>EC50: =2mg/L (48h, Daphnia magna) |

**Persistence and degradability** No information available.

**Bioaccumulation** There is no data for this product.

### Component Information

| Chemical name            | Partition coefficient |
|--------------------------|-----------------------|
| Ethyl alcohol<br>64-17-5 | -0.35                 |

**Other adverse effects** No information available.

## 13. Disposal considerations

### Disposal methods

**Waste from residues/unused products** Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation.

**Contaminated packaging** Do not reuse empty containers.

## 14. Transport information

**TDG** Not regulated

**DOT** Not regulated

**MEX** Not regulated

**IATA** Not regulated

**IMDG** Not regulated

**15. Regulatory information****Safety, health and environmental regulations/legislation specific for the substance or mixture****International Regulations**

The Montreal Protocol on Substances that Deplete the Ozone Layer Not applicable

The Stockholm Convention on Persistent Organic Pollutants Not applicable

The Rotterdam Convention Not applicable

**International Inventories**

Contact supplier for inventory compliance status

**16. Other information**

|             |                  |                |                    |                                    |
|-------------|------------------|----------------|--------------------|------------------------------------|
| <b>NFPA</b> | Health hazards 3 | Flammability 1 | Instability 0      | Physical and chemical properties - |
| <b>HMIS</b> | Health hazards 3 | Flammability 1 | Physical hazards 0 | Personal protection X              |

**Key or legend to abbreviations and acronyms used in the safety data sheet****Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | Sk*  | Skin designation                 |

**Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
 U.S. Environmental Protection Agency ChemView Database  
 European Food Safety Authority (EFSA)  
 Environmental Protection Agency  
 Acute Exposure Guideline Level(s) (AEGL(s))  
 U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
 U.S. Environmental Protection Agency High Production Volume Chemicals  
 Food Research Journal  
 Hazardous Substance Database  
 International Uniform Chemical Information Database (IUCLID)  
 Japan GHS Classification  
 Australia National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
 NIOSH (National Institute for Occupational Safety and Health)  
 National Library of Medicine's ChemID Plus (NLM CIP)  
 National Library of Medicine's PubMed database (NLM PUBMED)  
 U.S. National Toxicology Program (NTP)  
 New Zealand's Chemical Classification and Information Database (CCID)  
 Organization for Economic Co-operation and Development Environment, Health, and Safety Publications  
 Organization for Economic Co-operation and Development High Production Volume Chemicals Program  
 Organization for Economic Co-operation and Development Screening Information Data Set  
 RTECS (Registry of Toxic Effects of Chemical Substances)  
 World Health Organization

**Prepared By** Bio-Rad Laboratories, Environmental Health and Safety.

**Revision date** 06-Aug-2024

**Revision Note** Significant changes throughout SDS. Review all sections.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.



End of Safety Data Sheet



# SAFETY DATA SHEET

This safety data sheet was created pursuant to the requirements of:  
Canada WHMIS 2015 which includes the amended Hazardous Products Act (HPA) and the Hazardous Products Regulation (HPR)

Revision date 08-Aug-2024

Revision Number 1.3

## 1. Identification

### Product identifier

Product Name R1 - Microplate, 12 x 8 wells

### Other means of identification

Catalog Number(s) 7441A

### Recommended use of the chemical and restrictions on use

Recommended use Restricted to professional users  
In vitro diagnostic

Restrictions on use No information available

### Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer Address

Bio-Rad  
3 boulevard Raymond Poincaré  
92430 Marnes-la-Coquette  
France  
e-mail: fds-msds.fr@bio-rad.com

#### Legal Entity / Contact Address

Bio-Rad Laboratories (Canada) Ltd.  
2403 Guenette  
Montreal, Quebec H4R 2E9  
Canada

#### Technical Service

1-800-361-1808  
CSD\_Techsupport@bio-rad.com

### Emergency telephone number

24 Hour Emergency Phone Number CHEMTREC Canada:1 (800) 424-9300

## 2. Hazard(s) identification

### Classification

Not classified

### Label elements

#### Hazard statements

Not classified.

### Other information

No information available.

### 3. Composition/information on ingredients

#### Substance

Not applicable.

#### Mixture

The product contains no substances which at their given concentration, are considered to be hazardous to health

### 4. First-aid measures

#### Description of first aid measures

|                |                                                                                                                         |
|----------------|-------------------------------------------------------------------------------------------------------------------------|
| General advice | No hazards which require special first aid measures.                                                                    |
| Inhalation     | Remove to fresh air.                                                                                                    |
| Eye contact    | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids.<br>Consult a physician. |
| Skin contact   | Wash skin with soap and water.                                                                                          |
| Ingestion      | Rinse mouth thoroughly with water.                                                                                      |

#### Most important symptoms and effects, both acute and delayed

|          |                           |
|----------|---------------------------|
| Symptoms | No information available. |
|----------|---------------------------|

#### Indication of any immediate medical attention and special treatment needed

|                    |                        |
|--------------------|------------------------|
| Note to physicians | Treat symptomatically. |
|--------------------|------------------------|

### 5. Fire-fighting measures

|                                                |                                                                                                                                       |
|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Suitable Extinguishing Media                   | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.                               |
| Unsuitable extinguishing media                 | No information available.                                                                                                             |
| Specific hazards arising from the chemical     | None known.                                                                                                                           |
| Explosion data                                 |                                                                                                                                       |
| Sensitivity to mechanical impact               | None.                                                                                                                                 |
| Sensitivity to static discharge                | None.                                                                                                                                 |
| Special protective equipment for fire-fighters | Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear.<br>Use personal protection equipment. |

### 6. Accidental release measures

#### Personal precautions, protective equipment and emergency procedures

|                      |                                     |
|----------------------|-------------------------------------|
| Personal precautions | See section 8 for more information. |
|----------------------|-------------------------------------|

**Methods and material for containment and cleaning up**

|                         |                                                       |
|-------------------------|-------------------------------------------------------|
| Methods for containment | Prevent further leakage or spillage if safe to do so. |
| Methods for cleaning up | Pick up and transfer to properly labeled containers.  |

**7. Handling and storage****Precautions for safe handling**

|                         |                                                                        |
|-------------------------|------------------------------------------------------------------------|
| Advice on safe handling | Handle in accordance with good industrial hygiene and safety practice. |
|-------------------------|------------------------------------------------------------------------|

**Conditions for safe storage, including any incompatibilities**

|                    |                                                    |
|--------------------|----------------------------------------------------|
| Storage Conditions | Store according to product and label instructions. |
|--------------------|----------------------------------------------------|

**8. Exposure controls/personal protection****Control parameters**

|                 |                                                                                                                                                             |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Exposure Limits | This product, as supplied, does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies. |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|

**Appropriate engineering controls**

|                      |                                                     |
|----------------------|-----------------------------------------------------|
| Engineering controls | Showers<br>Eyewash stations<br>Ventilation systems. |
|----------------------|-----------------------------------------------------|

**Individual protection measures, such as personal protective equipment**

|                                |                                                                                                                                                                          |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Eye/face protection            | Wear safety glasses with side shields (or goggles).                                                                                                                      |
| Hand protection                | Wear suitable gloves.                                                                                                                                                    |
| Skin and body protection       | Wear suitable protective clothing.                                                                                                                                       |
| Respiratory protection         | No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required. |
| General hygiene considerations | Handle in accordance with good industrial hygiene and safety practice.                                                                                                   |

**9. Physical and chemical properties****Information on basic physical and chemical properties**

|                |                          |
|----------------|--------------------------|
| Physical state | Solid                    |
| Appearance     | solid                    |
| Color          | colorless                |
| Odor           | Odorless                 |
| Odor threshold | No information available |

| <u>Property</u>                | <u>Values</u>     | <u>Remarks • Method</u> |
|--------------------------------|-------------------|-------------------------|
| pH                             |                   | None known              |
| Melting point / freezing point | No data available | None known              |

|                                         |                    |            |
|-----------------------------------------|--------------------|------------|
| Initial boiling point and boiling range | No data available  | None known |
| Flash point                             | No data available  | None known |
| Evaporation rate                        | No data available  | None known |
| Flammability                            | No data available  | None known |
| Flammability Limit in Air               |                    | None known |
| Upper flammability or explosive limits  | No data available  |            |
| Lower flammability or explosive limits  | No data available  |            |
| Vapor pressure                          | No data available  | None known |
| Relative vapor density                  | No data available  | None known |
| Relative density                        | No data available  | None known |
| Water solubility                        | Insoluble in water |            |
| Solubility in other solvents            | No data available  | None known |
| Partition coefficient                   | No data available  | None known |
| Autoignition temperature                | No data available  | None known |
| Decomposition temperature               |                    | None known |
| Kinematic viscosity                     | No data available  | None known |
| Dynamic viscosity                       | No data available  | None known |
| <b>Other information</b>                |                    |            |
| Explosive properties                    | Not applicable.    |            |
| Oxidizing properties                    | Not applicable.    |            |
| Softening point                         | Not applicable     |            |
| Molecular weight                        | Not applicable     |            |
| VOC content                             | Not applicable     |            |

## 10. Stability and reactivity

|                                    |                                           |
|------------------------------------|-------------------------------------------|
| Reactivity                         | No information available.                 |
| Chemical stability                 | Stable under normal conditions.           |
| Possibility of hazardous reactions | None under normal processing.             |
| Conditions to avoid                | None known based on information supplied. |
| Incompatible materials             | None known based on information supplied. |
| Hazardous decomposition products   | None known based on information supplied. |

## 11. Toxicological information

### Information on likely routes of exposure

#### Product Information

|              |                                                                   |
|--------------|-------------------------------------------------------------------|
| Inhalation   | Specific test data for the substance or mixture is not available. |
| Eye contact  | Specific test data for the substance or mixture is not available. |
| Skin contact | Specific test data for the substance or mixture is not available. |
| Ingestion    | Specific test data for the substance or mixture is not available. |

### Symptoms related to the physical, chemical and toxicological characteristics

|          |                           |
|----------|---------------------------|
| Symptoms | No information available. |
|----------|---------------------------|

### Acute toxicity

#### Numerical measures of toxicity

No information available

**Delayed and immediate effects as well as chronic effects from short and long-term exposure**

|                                          |                                                                   |
|------------------------------------------|-------------------------------------------------------------------|
| <b>Skin corrosion/irritation</b>         | Based on available data, the classification criteria are not met. |
| <b>Serious eye damage/eye irritation</b> | Based on available data, the classification criteria are not met. |
| <b>Respiratory or skin sensitization</b> | Based on available data, the classification criteria are not met. |
| <b>Germ cell mutagenicity</b>            | Based on available data, the classification criteria are not met. |
| <b>Carcinogenicity</b>                   | Based on available data, the classification criteria are not met. |
| <b>Reproductive toxicity</b>             | Based on available data, the classification criteria are not met. |
| <b>STOT - single exposure</b>            | Based on available data, the classification criteria are not met. |
| <b>STOT - repeated exposure</b>          | Based on available data, the classification criteria are not met. |
| <b>Aspiration hazard</b>                 | Based on available data, the classification criteria are not met. |

**12. Ecological information**

|                                      |                                                                           |
|--------------------------------------|---------------------------------------------------------------------------|
| <b>Ecotoxicity</b>                   | The environmental impact of this product has not been fully investigated. |
| <b>Persistence and degradability</b> | No information available.                                                 |
| <b>Bioaccumulation</b>               | No information available.                                                 |
| <b>Other adverse effects</b>         | No information available.                                                 |

**13. Disposal considerations****Disposal methods**

|                                            |                                                                                                                 |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>Waste from residues/unused products</b> | Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation. |
| <b>Contaminated packaging</b>              | Do not reuse empty containers.                                                                                  |

**14. Transport information**

|                    |               |
|--------------------|---------------|
| <b><u>TDG</u></b>  | Not regulated |
| <b><u>DOT</u></b>  | Not regulated |
| <b><u>MEX</u></b>  | Not regulated |
| <b><u>IATA</u></b> | Not regulated |
| <b><u>IMDG</u></b> | Not regulated |

**15. Regulatory information****Safety, health and environmental regulations/legislation specific for the substance or mixture****International Regulations****The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable**The Stockholm Convention on Persistent Organic Pollutants** Not applicable**The Rotterdam Convention** Not applicable**International Inventories**

Contact supplier for inventory compliance status

**16. Other information**

|             |                         |                       |                           |                                           |
|-------------|-------------------------|-----------------------|---------------------------|-------------------------------------------|
| <b>NFPA</b> | <b>Health hazards</b> 0 | <b>Flammability</b> 0 | <b>Instability</b> 0      | <b>Physical and chemical properties -</b> |
| <b>HMIS</b> | <b>Health hazards</b> 0 | <b>Flammability</b> 0 | <b>Physical hazards</b> 0 | <b>Personal protection</b> X              |

**Key or legend to abbreviations and acronyms used in the safety data sheet****Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | Sk*  | Skin designation                 |

**Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
U.S. Environmental Protection Agency ChemView Database  
European Food Safety Authority (EFSA)  
Environmental Protection Agency  
Acute Exposure Guideline Level(s) (AEGL(s))  
U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
U.S. Environmental Protection Agency High Production Volume Chemicals  
Food Research Journal  
Hazardous Substance Database  
International Uniform Chemical Information Database (IUCLID)  
Japan GHS Classification  
Australia National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
NIOSH (National Institute for Occupational Safety and Health)  
National Library of Medicine's ChemID Plus (NLM CIP)  
National Library of Medicine's PubMed database (NLM PUBMED)  
U.S. National Toxicology Program (NTP)  
New Zealand's Chemical Classification and Information Database (CCID)  
Organization for Economic Co-operation and Development Environment, Health, and Safety Publications  
Organization for Economic Co-operation and Development High Production Volume Chemicals Program  
Organization for Economic Co-operation and Development Screening Information Data Set  
RTECS (Registry of Toxic Effects of Chemical Substances)  
World Health Organization

**Prepared By** Bio-Rad Laboratories, Environmental Health and Safety.**Revision date** 08-Aug-2024**Revision Note** Significant changes throughout SDS. Review all sections.**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information

relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

**End of Safety Data Sheet**





# SAFETY DATA SHEET

This safety data sheet was created pursuant to the requirements of:  
Canada WHMIS 2015 which includes the amended Hazardous Products Act (HPA) and the Hazardous Products Regulation (HPR)

Revision date 08-Aug-2024

Revision Number 1.2

## 1. Identification

### Product identifier

Product Name C0 - Negative Control, 2.2 ml

### Other means of identification

Catalog Number(s) 7441C

### Recommended use of the chemical and restrictions on use

Recommended use Restricted to professional users  
In vitro diagnostic

Restrictions on use No information available

### Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer Address

Bio-Rad  
3 boulevard Raymond Poincaré  
92430 Marnes-la-Coquette  
France  
e-mail: fds-msds.fr@bio-rad.com

#### Legal Entity / Contact Address

Bio-Rad Laboratories (Canada) Ltd.  
2403 Guenette  
Montreal, Quebec H4R 2E9  
Canada

#### Technical Service

1-800-361-1808  
CSD\_Techsupport@bio-rad.com

### Emergency telephone number

24 Hour Emergency Phone Number CHEMTREC Canada:1 (800) 424-9300

## 2. Hazard(s) identification

### Classification

Skin sensitization

Category 1

### Label elements

#### **Warning**

#### **Hazard statements**

May cause an allergic skin reaction



**Precautionary Statements - Prevention**

Avoid breathing dust/fume/gas/mist/vapors/spray  
Contaminated work clothing should not be allowed out of the workplace  
Wear protective gloves/protective clothing/eye protection/face protection

**Skin**

IF ON SKIN: Wash with plenty of water and soap  
If skin irritation or rash occurs: Get medical advice/attention  
Take off contaminated clothing and wash it before reuse

**Precautionary Statements - Disposal**

Dispose of contents/container in accordance with local, regional, national, and international regulations as applicable

**Other information**

Contains animal source material. (Cattle).

**3. Composition/information on ingredients****Substance**

Not applicable.

**Mixture**

The product contains no substances which at their given concentration, are considered to be hazardous to health

| Chemical name                                                                       | CAS No.    | Weight-%     | Hazardous Material<br>Information Review Act<br>registry number<br>(HMIRA registry #) | Date HMIRA filed and<br>date exemption granted<br>(if applicable) |
|-------------------------------------------------------------------------------------|------------|--------------|---------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with<br>2-methyl-3(2H)-isothiazolone | 55965-84-9 | 0.01 - 0.099 | -                                                                                     |                                                                   |

**4. First-aid measures****Description of first aid measures**

|                       |                                                                                                                                         |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>General advice</b> | Show this safety data sheet to the doctor in attendance.                                                                                |
| <b>Inhalation</b>     | Remove to fresh air.                                                                                                                    |
| <b>Eye contact</b>    | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids.<br>Consult a physician.                 |
| <b>Skin contact</b>   | Wash with soap and water. May cause an allergic skin reaction. In the case of skin irritation<br>or allergic reactions see a physician. |
| <b>Ingestion</b>      | Rinse mouth thoroughly with water.                                                                                                      |

**Most important symptoms and effects, both acute and delayed**

|                 |                         |
|-----------------|-------------------------|
| <b>Symptoms</b> | Itching. Rashes. Hives. |
|-----------------|-------------------------|

**Indication of any immediate medical attention and special treatment needed**

**Note to physicians** May cause sensitization in susceptible persons. Treat symptomatically.

## 5. Fire-fighting measures

|                                                       |                                                                                                                                    |
|-------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>                   | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.                            |
| <b>Unsuitable extinguishing media</b>                 | No information available.                                                                                                          |
| <b>Specific hazards arising from the chemical</b>     | Product is or contains a sensitizer. May cause sensitization by skin contact.                                                      |
| <b>Explosion data</b>                                 |                                                                                                                                    |
| <b>Sensitivity to mechanical impact</b>               | None.                                                                                                                              |
| <b>Sensitivity to static discharge</b>                | None.                                                                                                                              |
| <b>Special protective equipment for fire-fighters</b> | Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment. |

## 6. Accidental release measures

### Personal precautions, protective equipment and emergency procedures

|                             |                                                                                                                                                                                                          |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal precautions</b> | Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Methods and material for containment and cleaning up

|                                |                                                       |
|--------------------------------|-------------------------------------------------------|
| <b>Methods for containment</b> | Prevent further leakage or spillage if safe to do so. |
| <b>Methods for cleaning up</b> | Pick up and transfer to properly labeled containers.  |

## 7. Handling and storage

### Precautions for safe handling

|                                |                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Advice on safe handling</b> | Handle in accordance with good industrial hygiene and safety practice. Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. In case of insufficient ventilation, wear suitable respiratory equipment. Do not eat, drink or smoke when using this product. Take off contaminated clothing and wash before reuse. |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Conditions for safe storage, including any incompatibilities

|                           |                                                                                                                                                                                 |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Storage Conditions</b> | Keep containers tightly closed in a dry, cool and well-ventilated place. Store locked up. Keep out of the reach of children. Store according to product and label instructions. |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 8. Exposure controls/personal protection

### Control parameters

|                        |                                                                                                                                                             |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Exposure Limits</b> | This product, as supplied, does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies. |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Appropriate engineering controls

## 9. Physical and chemical properties

|                       |                          |
|-----------------------|--------------------------|
| <b>Physical state</b> | Liquid                   |
| <b>Appearance</b>     | Liquid                   |
| <b>Color</b>          | light yellow             |
| <b>Odor</b>           | Odorless                 |
| <b>Odor threshold</b> | No information available |

## 10. Stability and reactivity

|                   |                           |
|-------------------|---------------------------|
| <b>Reactivity</b> | No information available. |
|-------------------|---------------------------|

|                                    |                                           |
|------------------------------------|-------------------------------------------|
| Chemical stability                 | Stable under normal conditions.           |
| Possibility of hazardous reactions | None under normal processing.             |
| Conditions to avoid                | None known based on information supplied. |
| Incompatible materials             | None known based on information supplied. |
| Hazardous decomposition products   | None known based on information supplied. |

## 11. Toxicological information

### Information on likely routes of exposure

#### Product Information

|              |                                                                                                                                                                                                                           |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Inhalation   | Specific test data for the substance or mixture is not available.                                                                                                                                                         |
| Eye contact  | Specific test data for the substance or mixture is not available.                                                                                                                                                         |
| Skin contact | May cause sensitization by skin contact. Specific test data for the substance or mixture is not available Repeated or prolonged skin contact may cause allergic reactions with susceptible persons (based on components). |
| Ingestion    | Specific test data for the substance or mixture is not available.                                                                                                                                                         |

### Symptoms related to the physical, chemical and toxicological characteristics

|          |                         |
|----------|-------------------------|
| Symptoms | Itching. Rashes. Hives. |
|----------|-------------------------|

### Acute toxicity

#### Numerical measures of toxicity

No information available

#### Component Information

| Chemical name                                                                                  | Oral LD50          | Dermal LD50              | Inhalation LC50 |
|------------------------------------------------------------------------------------------------|--------------------|--------------------------|-----------------|
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with 2-methyl-3(2H)-isothiazolone<br>55965-84-9 | = 53 mg/kg ( Rat ) | = 87.12 mg/kg ( Rabbit ) | -               |

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                                   |                                                                   |
|-----------------------------------|-------------------------------------------------------------------|
| Skin corrosion/irritation         | Based on available data, the classification criteria are not met. |
| Serious eye damage/eye irritation | Based on available data, the classification criteria are not met. |
| Respiratory or skin sensitization | May cause sensitization by skin contact.                          |
| Germ cell mutagenicity            | Based on available data, the classification criteria are not met. |
| Carcinogenicity                   | Based on available data, the classification criteria are not met. |
| Reproductive toxicity             | Based on available data, the classification criteria are not met. |
| STOT - single exposure            | Based on available data, the classification criteria are not met. |
| STOT - repeated exposure          | Based on available data, the classification criteria are not met. |

**Aspiration hazard** Based on available data, the classification criteria are not met.

## 12. Ecological information

**Ecotoxicity** Harmful to aquatic life with long lasting effects.

**Persistence and degradability** No information available.

**Bioaccumulation** There is no data for this product.

### Component Information

| Chemical name                                                                                     | Partition coefficient |
|---------------------------------------------------------------------------------------------------|-----------------------|
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with<br>2-methyl-3(2H)-isothiazolone<br>55965-84-9 | 0.7                   |

**Other adverse effects** No information available.

## 13. Disposal considerations

### Disposal methods

**Waste from residues/unused products** Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation.

**Contaminated packaging** Do not reuse empty containers.

## 14. Transport information

**TDG** Not regulated

**DOT** Not regulated

**MEX** Not regulated

**IATA** Not regulated

**IMDG** Not regulated

## 15. Regulatory information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Regulations

**The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable

**The Stockholm Convention on Persistent Organic Pollutants** Not applicable

**The Rotterdam Convention** Not applicable

#### International Inventories

Contact supplier for inventory compliance status

**16. Other information**

|             |                         |                       |                           |                                           |
|-------------|-------------------------|-----------------------|---------------------------|-------------------------------------------|
| <b>NFPA</b> | <b>Health hazards</b> 2 | <b>Flammability</b> 0 | <b>Instability</b> 0      | <b>Physical and chemical properties</b> - |
| <b>HMIS</b> | <b>Health hazards</b> 2 | <b>Flammability</b> 0 | <b>Physical hazards</b> 0 | <b>Personal protection</b> X              |

**Key or legend to abbreviations and acronyms used in the safety data sheet****Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | Sk*  | Skin designation                 |

**Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
U.S. Environmental Protection Agency ChemView Database  
European Food Safety Authority (EFSA)  
Environmental Protection Agency  
Acute Exposure Guideline Level(s) (AEGL(s))  
U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
U.S. Environmental Protection Agency High Production Volume Chemicals  
Food Research Journal  
Hazardous Substance Database  
International Uniform Chemical Information Database (IUCLID)  
Japan GHS Classification  
Australia National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
NIOSH (National Institute for Occupational Safety and Health)  
National Library of Medicine's ChemID Plus (NLM CIP)  
National Library of Medicine's PubMed database (NLM PUBMED)  
U.S. National Toxicology Program (NTP)  
New Zealand's Chemical Classification and Information Database (CCID)  
Organization for Economic Co-operation and Development Environment, Health, and Safety Publications  
Organization for Economic Co-operation and Development High Production Volume Chemicals Program  
Organization for Economic Co-operation and Development Screening Information Data Set  
RTECS (Registry of Toxic Effects of Chemical Substances)  
World Health Organization

**Prepared By** Bio-Rad Laboratories, Environmental Health and Safety.

**Revision date** 08-Aug-2024

**Revision Note** Significant changes throughout SDS. Review all sections.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

**End of Safety Data Sheet**



# SAFETY DATA SHEET

This safety data sheet was created pursuant to the requirements of:  
Canada WHMIS 2015 which includes the amended Hazardous Products Act (HPA) and the Hazardous Products Regulation (HPR)

Revision date 07-Aug-2024

Revision Number 1.1

## 1. Identification

### Product identifier

Product Name C1 - 10 mIU/ml Calibrator, 3 ml

### Other means of identification

Catalog Number(s) 7441E

### Recommended use of the chemical and restrictions on use

Recommended use Restricted to professional users  
In vitro diagnostic

Restrictions on use No information available

### Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer Address

Bio-Rad  
3 boulevard Raymond Poincaré  
92430 Marnes-la-Coquette  
France  
e-mail: fds-msds.fr@bio-rad.com

#### Legal Entity / Contact Address

Bio-Rad Laboratories (Canada) Ltd.  
2403 Guenette  
Montreal, Quebec H4R 2E9  
Canada

Technical Service 1-800-361-1808  
CSD\_Techsupport@bio-rad.com

### Emergency telephone number

24 Hour Emergency Phone Number CHEMTREC Canada:1 (800) 424-9300

## 2. Hazard(s) identification

### Classification

Skin sensitization

Category 1

### Label elements

#### **Warning**

#### **Hazard statements**

May cause an allergic skin reaction





**Precautionary Statements - Prevention**

Avoid breathing dust/fume/gas/mist/vapors/spray  
Contaminated work clothing should not be allowed out of the workplace  
Wear protective gloves/protective clothing/eye protection/face protection

**Skin**

IF ON SKIN: Wash with plenty of water and soap  
If skin irritation or rash occurs: Get medical advice/attention  
Take off contaminated clothing and wash it before reuse

**Precautionary Statements - Disposal**

Dispose of contents/container in accordance with local, regional, national, and international regulations as applicable

**Other information**

Contains animal source material. (Cattle).  
Contains human source material and / or potentially infectious components

**3. Composition/information on ingredients****Substance**

Not applicable.

**Mixture**

The product contains no substances which at their given concentration, are considered to be hazardous to health

| Chemical name                                                                       | CAS No.    | Weight-%     | Hazardous Material<br>Information Review Act<br>registry number<br>(HMIRA registry #) | Date HMIRA filed and<br>date exemption granted<br>(if applicable) |
|-------------------------------------------------------------------------------------|------------|--------------|---------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with<br>2-methyl-3(2H)-isothiazolone | 55965-84-9 | 0.01 - 0.099 | -                                                                                     |                                                                   |

**4. First-aid measures****Description of first aid measures**

|                       |                                                                                                                                                                                                 |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General advice</b> | Show this safety data sheet to the doctor in attendance.                                                                                                                                        |
| <b>Inhalation</b>     | Remove to fresh air.                                                                                                                                                                            |
| <b>Eye contact</b>    | Contains human source material and / or potentially infectious components. Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids. Consult a physician. |
| <b>Skin contact</b>   | Wash with soap and water. May cause an allergic skin reaction. In the case of skin irritation or allergic reactions see a physician. Wash skin with soap and water.                             |
| <b>Ingestion</b>      | Call a physician. Contains human source material and / or potentially infectious components.                                                                                                    |

**Most important symptoms and effects, both acute and delayed**

**Symptoms** Itching. Rashes. Hives.

**Indication of any immediate medical attention and special treatment needed**

**Note to physicians** May cause sensitization in susceptible persons. Treat symptomatically. Contains human source material and / or potentially infectious components.

## 5. Fire-fighting measures

**Suitable Extinguishing Media** Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

**Unsuitable extinguishing media** No information available.

**Specific hazards arising from the chemical** Product is or contains a sensitizer. May cause sensitization by skin contact.

**Explosion data**

**Sensitivity to mechanical impact** None.

**Sensitivity to static discharge** None.

**Special protective equipment for fire-fighters** Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment.

## 6. Accidental release measures

**Personal precautions, protective equipment and emergency procedures**

**Personal precautions** Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak.

**Methods and material for containment and cleaning up**

**Methods for containment** Do not allow into any sewer, on the ground or into any body of water.

**Methods for cleaning up** Clean contaminated surface thoroughly. Use: Disinfectant.

## 7. Handling and storage

**Precautions for safe handling**

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice. Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. In case of insufficient ventilation, wear suitable respiratory equipment. Do not eat, drink or smoke when using this product. Take off contaminated clothing and wash before reuse.

**Conditions for safe storage, including any incompatibilities**

**Storage Conditions** Keep containers tightly closed in a dry, cool and well-ventilated place. Store locked up. Keep out of the reach of children. Store according to product and label instructions.

## 8. Exposure controls/personal protection

**Control parameters**

**Exposure Limits** This product, as supplied, does not contain any hazardous materials with occupational

exposure limits established by the region specific regulatory bodies.

### Appropriate engineering controls

|                             |                      |
|-----------------------------|----------------------|
| <b>Engineering controls</b> | Showers              |
|                             | Eyewash stations     |
|                             | Ventilation systems. |

### Individual protection measures, such as personal protective equipment

|                                       |                                                                                                                                                                          |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye/face protection</b>            | Wear safety glasses with side shields (or goggles).                                                                                                                      |
| <b>Hand protection</b>                | Wear suitable gloves.                                                                                                                                                    |
| <b>Skin and body protection</b>       | Wear suitable protective clothing.                                                                                                                                       |
| <b>Respiratory protection</b>         | No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required. |
| <b>General hygiene considerations</b> | Follow universal and standard precautions for handling potentially infectious materials.                                                                                 |

## 9. Physical and chemical properties

### Information on basic physical and chemical properties

|                       |                          |
|-----------------------|--------------------------|
| <b>Physical state</b> | Liquid                   |
| <b>Appearance</b>     | Liquid                   |
| <b>Color</b>          | yellow                   |
| <b>Odor</b>           | Odorless                 |
| <b>Odor threshold</b> | No information available |

| <u>Property</u>                         | <u>Values</u>     | <u>Remarks • Method</u> |
|-----------------------------------------|-------------------|-------------------------|
| pH                                      |                   | None known              |
| Melting point / freezing point          | No data available | None known              |
| Initial boiling point and boiling range | No data available | None known              |
| Flash point                             | No data available | None known              |
| Evaporation rate                        | No data available | None known              |
| Flammability                            | No data available | None known              |
| Flammability Limit in Air               |                   | None known              |
| Upper flammability or explosive limits  | No data available |                         |
| Lower flammability or explosive limits  | No data available |                         |
| Vapor pressure                          | No data available | None known              |
| Relative vapor density                  | No data available | None known              |
| Relative density                        | No data available | None known              |
| Water solubility                        | Miscible in water |                         |
| Solubility in other solvents            | No data available | None known              |
| Partition coefficient                   | No data available | None known              |
| Autoignition temperature                | No data available | None known              |
| Decomposition temperature               |                   | None known              |
| Kinematic viscosity                     | No data available | None known              |
| Dynamic viscosity                       | No data available | None known              |

## Other information

|                      |                 |
|----------------------|-----------------|
| Explosive properties | Not applicable. |
| Oxidizing properties | Not applicable. |
| Softening point      | Not applicable  |
| Molecular weight     | Not applicable  |
| VOC content          | Not applicable  |

## 10. Stability and reactivity

|                                    |                                           |
|------------------------------------|-------------------------------------------|
| Reactivity                         | No information available.                 |
| Chemical stability                 | Stable under normal conditions.           |
| Possibility of hazardous reactions | None under normal processing.             |
| Conditions to avoid                | None known based on information supplied. |
| Incompatible materials             | None known based on information supplied. |
| Hazardous decomposition products   | None known based on information supplied. |

## 11. Toxicological information

### Information on likely routes of exposure

#### Product Information

|              |                                                                                                                                                                                                                           |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Inhalation   | Specific test data for the substance or mixture is not available.                                                                                                                                                         |
| Eye contact  | Specific test data for the substance or mixture is not available.                                                                                                                                                         |
| Skin contact | May cause sensitization by skin contact. Specific test data for the substance or mixture is not available Repeated or prolonged skin contact may cause allergic reactions with susceptible persons (based on components). |
| Ingestion    | Specific test data for the substance or mixture is not available.                                                                                                                                                         |

### Symptoms related to the physical, chemical and toxicological characteristics

|          |                         |
|----------|-------------------------|
| Symptoms | Itching. Rashes. Hives. |
|----------|-------------------------|

### Acute toxicity

#### Numerical measures of toxicity

No information available

#### Component Information

| Chemical name                                                                                  | Oral LD50          | Dermal LD50              | Inhalation LC50 |
|------------------------------------------------------------------------------------------------|--------------------|--------------------------|-----------------|
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with 2-methyl-3(2H)-isothiazolone<br>55965-84-9 | = 53 mg/kg ( Rat ) | = 87.12 mg/kg ( Rabbit ) | -               |

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                                   |                                                                   |
|-----------------------------------|-------------------------------------------------------------------|
| Skin corrosion/irritation         | Based on available data, the classification criteria are not met. |
| Serious eye damage/eye irritation | Based on available data, the classification criteria are not met. |
| Respiratory or skin sensitization | May cause sensitization by skin contact.                          |
| Germ cell mutagenicity            | Based on available data, the classification criteria are not met. |
| Carcinogenicity                   | Based on available data, the classification criteria are not met. |
| Reproductive toxicity             | Based on available data, the classification criteria are not met. |

|                          |                                                                   |
|--------------------------|-------------------------------------------------------------------|
| STOT - single exposure   | Based on available data, the classification criteria are not met. |
| STOT - repeated exposure | Based on available data, the classification criteria are not met. |
| Aspiration hazard        | Based on available data, the classification criteria are not met. |

## 12. Ecological information

**Ecotoxicity** Harmful to aquatic life with long lasting effects.

**Persistence and degradability** No information available.

**Bioaccumulation** There is no data for this product.

### Component Information

| Chemical name                                                                                     | Partition coefficient |
|---------------------------------------------------------------------------------------------------|-----------------------|
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with<br>2-methyl-3(2H)-isothiazolone<br>55965-84-9 | 0.7                   |

**Other adverse effects** No information available.

## 13. Disposal considerations

### Disposal methods

**Waste from residues/unused products** Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation.

**Contaminated packaging** Do not reuse empty containers.

## 14. Transport information

**TDG** Not regulated

**DOT** Not regulated

**MEX** Not regulated

**IATA** Not regulated

**IMDG** Not regulated

## 15. Regulatory information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Regulations

**The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable

**The Stockholm Convention on Persistent Organic Pollutants** Not applicable

**The Rotterdam Convention** Not applicable

**International Inventories**

Contact supplier for inventory compliance status

**16. Other information**

|             |                         |                       |                           |                                           |
|-------------|-------------------------|-----------------------|---------------------------|-------------------------------------------|
| <b>NFPA</b> | <b>Health hazards</b> 2 | <b>Flammability</b> 0 | <b>Instability</b> 0      | <b>Physical and chemical properties -</b> |
| <b>HMIS</b> | <b>Health hazards</b> 2 | <b>Flammability</b> 0 | <b>Physical hazards</b> 0 | <b>Personal protection</b> X              |

**Key or legend to abbreviations and acronyms used in the safety data sheet****Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | Sk*  | Skin designation                 |

**Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
U.S. Environmental Protection Agency ChemView Database  
European Food Safety Authority (EFSA)  
Environmental Protection Agency  
Acute Exposure Guideline Level(s) (AEGL(s))  
U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
U.S. Environmental Protection Agency High Production Volume Chemicals  
Food Research Journal  
Hazardous Substance Database  
International Uniform Chemical Information Database (IUCLID)  
Japan GHS Classification  
Australia National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
NIOSH (National Institute for Occupational Safety and Health)  
National Library of Medicine's ChemID Plus (NLM CIP)  
National Library of Medicine's PubMed database (NLM PUBMED)  
U.S. National Toxicology Program (NTP)  
New Zealand's Chemical Classification and Information Database (CCID)  
Organization for Economic Co-operation and Development Environment, Health, and Safety Publications  
Organization for Economic Co-operation and Development High Production Volume Chemicals Program  
Organization for Economic Co-operation and Development Screening Information Data Set  
RTECS (Registry of Toxic Effects of Chemical Substances)  
World Health Organization

**Prepared By** Bio-Rad Laboratories, Environmental Health and Safety.

**Revision date** 07-Aug-2024

**Revision Note** Significant changes throughout SDS. Review all sections.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

**End of Safety Data Sheet**



# SAFETY DATA SHEET

This safety data sheet was created pursuant to the requirements of:  
Canada WHMIS 2015 which includes the amended Hazardous Products Act (HPA) and the Hazardous Products Regulation (HPR)

Revision date 21-Aug-2024

Revision Number 1.1

## 1. Identification

### Product identifier

**Product Name** C2 - 100 mIU/ml Calibrator / Positive Control, 2.2 ml

### Other means of identification

**Catalog Number(s)** 7441D

### Recommended use of the chemical and restrictions on use

**Recommended use** Restricted to professional users  
In vitro diagnostic

**Restrictions on use** No information available

### Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer Address

Bio-Rad  
3 boulevard Raymond Poincaré  
92430 Marnes-la-Coquette  
France  
e-mail: fds-msds.fr@bio-rad.com

#### Legal Entity / Contact Address

Bio-Rad Laboratories (Canada) Ltd.  
2403 Guenette  
Montreal, Quebec H4R 2E9  
Canada

**Technical Service** 1-800-361-1808  
CSD\_Techsupport@bio-rad.com

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC Canada:1 (800) 424-9300

## 2. Hazard(s) identification

### Classification

|                    |            |
|--------------------|------------|
| Skin sensitization | Category 1 |
|--------------------|------------|

### Label elements

#### **Warning**

#### **Hazard statements**

May cause an allergic skin reaction



**Precautionary Statements - Prevention**

Avoid breathing dust/fume/gas/mist/vapors/spray  
Contaminated work clothing should not be allowed out of the workplace  
Wear protective gloves/protective clothing/eye protection/face protection

**Skin**

IF ON SKIN: Wash with plenty of water and soap  
If skin irritation or rash occurs: Get medical advice/attention  
Take off contaminated clothing and wash it before reuse

**Precautionary Statements - Disposal**

Dispose of contents/container in accordance with local, regional, national, and international regulations as applicable

**Other information**

Contains animal source material. (Cattle).  
Contains human source material and / or potentially infectious components

**3. Composition/information on ingredients****Substance**

Not applicable.

**Mixture**

The product contains no substances which at their given concentration, are considered to be hazardous to health

| Chemical name                                                                       | CAS No.    | Weight-%     | Hazardous Material<br>Information Review Act<br>registry number<br>(HMIRA registry #) | Date HMIRA filed and<br>date exemption granted<br>(if applicable) |
|-------------------------------------------------------------------------------------|------------|--------------|---------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with<br>2-methyl-3(2H)-isothiazolone | 55965-84-9 | 0.01 - 0.099 | -                                                                                     |                                                                   |

**4. First-aid measures****Description of first aid measures**

|                       |                                                                                                                                                                                                 |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General advice</b> | Show this safety data sheet to the doctor in attendance.                                                                                                                                        |
| <b>Inhalation</b>     | Remove to fresh air.                                                                                                                                                                            |
| <b>Eye contact</b>    | Contains human source material and / or potentially infectious components. Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids. Consult a physician. |
| <b>Skin contact</b>   | Wash with soap and water. May cause an allergic skin reaction. In the case of skin irritation or allergic reactions see a physician. Wash skin with soap and water.                             |
| <b>Ingestion</b>      | Call a physician. Contains human source material and / or potentially infectious components.                                                                                                    |

**Most important symptoms and effects, both acute and delayed**



**Symptoms** Itching. Rashes. Hives.

**Indication of any immediate medical attention and special treatment needed**

**Note to physicians** May cause sensitization in susceptible persons. Treat symptomatically. Contains human source material and / or potentially infectious components.

## 5. Fire-fighting measures

**Suitable Extinguishing Media** Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

**Unsuitable extinguishing media** No information available.

**Specific hazards arising from the chemical** Product is or contains a sensitizer. May cause sensitization by skin contact.

**Explosion data**

**Sensitivity to mechanical impact** None.

**Sensitivity to static discharge** None.

**Special protective equipment for fire-fighters** Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment.

## 6. Accidental release measures

**Personal precautions, protective equipment and emergency procedures**

**Personal precautions** Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak.

**Methods and material for containment and cleaning up**

**Methods for containment** Do not allow into any sewer, on the ground or into any body of water.

**Methods for cleaning up** Clean contaminated surface thoroughly. Use: Disinfectant.

## 7. Handling and storage

**Precautions for safe handling**

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice. Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. In case of insufficient ventilation, wear suitable respiratory equipment. Do not eat, drink or smoke when using this product. Take off contaminated clothing and wash before reuse.

**Conditions for safe storage, including any incompatibilities**

**Storage Conditions** Keep containers tightly closed in a dry, cool and well-ventilated place. Store locked up. Keep out of the reach of children. Store according to product and label instructions.

## 8. Exposure controls/personal protection

**Control parameters**

**Exposure Limits** This product, as supplied, does not contain any hazardous materials with occupational

exposure limits established by the region specific regulatory bodies.

#### Appropriate engineering controls

**Engineering controls**                      Showers  
                                                          Eyewash stations  
                                                          Ventilation systems.

#### Individual protection measures, such as personal protective equipment

**Eye/face protection**                      Wear safety glasses with side shields (or goggles).

**Hand protection**                              Wear suitable gloves.

**Skin and body protection**                      Wear suitable protective clothing.

**Respiratory protection**                      No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

**General hygiene considerations**                      Follow universal and standard precautions for handling potentially infectious materials.

## 9. Physical and chemical properties

#### Information on basic physical and chemical properties

**Physical state**                                      Liquid  
**Appearance**                                      Liquid  
**Color**                                                  light green  
**Odor**                                                  Odorless  
**Odor threshold**                                      No information available

| <u>Property</u>                                | <u>Values</u>     | <u>Remarks • Method</u> |
|------------------------------------------------|-------------------|-------------------------|
| <b>pH</b>                                      |                   | None known              |
| <b>Melting point / freezing point</b>          | No data available | None known              |
| <b>Initial boiling point and boiling range</b> | No data available | None known              |
| <b>Flash point</b>                             | No data available | None known              |
| <b>Evaporation rate</b>                        | No data available | None known              |
| <b>Flammability</b>                            | No data available | None known              |
| <b>Flammability Limit in Air</b>               |                   | None known              |
| <b>Upper flammability or explosive limits</b>  | No data available |                         |
| <b>Lower flammability or explosive limits</b>  | No data available |                         |
| <b>Vapor pressure</b>                          | No data available | None known              |
| <b>Relative vapor density</b>                  | No data available | None known              |
| <b>Relative density</b>                        | No data available | None known              |
| <b>Water solubility</b>                        | Miscible in water |                         |
| <b>Solubility in other solvents</b>            | No data available | None known              |
| <b>Partition coefficient</b>                   | No data available | None known              |
| <b>Autoignition temperature</b>                | No data available | None known              |
| <b>Decomposition temperature</b>               |                   | None known              |
| <b>Kinematic viscosity</b>                     | No data available | None known              |
| <b>Dynamic viscosity</b>                       | No data available | None known              |

#### Other information

**Explosive properties**                              Not applicable.  
**Oxidizing properties**                              Not applicable.  
**Softening point**                                      Not applicable  
**Molecular weight**                                      Not applicable  
**VOC content**                                          Not applicable

**10. Stability and reactivity**

|                                    |                                           |
|------------------------------------|-------------------------------------------|
| Reactivity                         | No information available.                 |
| Chemical stability                 | Stable under normal conditions.           |
| Possibility of hazardous reactions | None under normal processing.             |
| Conditions to avoid                | None known based on information supplied. |
| Incompatible materials             | None known based on information supplied. |
| Hazardous decomposition products   | None known based on information supplied. |

**11. Toxicological information****Information on likely routes of exposure****Product Information**

|              |                                                                                                                                                                                                                           |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Inhalation   | Specific test data for the substance or mixture is not available.                                                                                                                                                         |
| Eye contact  | Specific test data for the substance or mixture is not available.                                                                                                                                                         |
| Skin contact | May cause sensitization by skin contact. Specific test data for the substance or mixture is not available Repeated or prolonged skin contact may cause allergic reactions with susceptible persons (based on components). |
| Ingestion    | Specific test data for the substance or mixture is not available.                                                                                                                                                         |

**Symptoms related to the physical, chemical and toxicological characteristics**

|          |                         |
|----------|-------------------------|
| Symptoms | Itching. Rashes. Hives. |
|----------|-------------------------|

**Acute toxicity**

**Numerical measures of toxicity**  
No information available

**Component Information**

| Chemical name                                                                                  | Oral LD50          | Dermal LD50              | Inhalation LC50 |
|------------------------------------------------------------------------------------------------|--------------------|--------------------------|-----------------|
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with 2-methyl-3(2H)-isothiazolone<br>55965-84-9 | = 53 mg/kg ( Rat ) | = 87.12 mg/kg ( Rabbit ) | -               |

**Delayed and immediate effects as well as chronic effects from short and long-term exposure**

|                                   |                                                                   |
|-----------------------------------|-------------------------------------------------------------------|
| Skin corrosion/irritation         | Based on available data, the classification criteria are not met. |
| Serious eye damage/eye irritation | Based on available data, the classification criteria are not met. |
| Respiratory or skin sensitization | May cause sensitization by skin contact.                          |
| Germ cell mutagenicity            | Based on available data, the classification criteria are not met. |
| Carcinogenicity                   | Based on available data, the classification criteria are not met. |
| Reproductive toxicity             | Based on available data, the classification criteria are not met. |

|                          |                                                                   |
|--------------------------|-------------------------------------------------------------------|
| STOT - single exposure   | Based on available data, the classification criteria are not met. |
| STOT - repeated exposure | Based on available data, the classification criteria are not met. |
| Aspiration hazard        | Based on available data, the classification criteria are not met. |

## 12. Ecological information

|                               |                                                    |
|-------------------------------|----------------------------------------------------|
| Ecotoxicity                   | Harmful to aquatic life with long lasting effects. |
| Persistence and degradability | No information available.                          |
| Bioaccumulation               | There is no data for this product.                 |

### Component Information

| Chemical name                                                                                     | Partition coefficient |
|---------------------------------------------------------------------------------------------------|-----------------------|
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with<br>2-methyl-3(2H)-isothiazolone<br>55965-84-9 | 0.7                   |

|                       |                           |
|-----------------------|---------------------------|
| Other adverse effects | No information available. |
|-----------------------|---------------------------|

## 13. Disposal considerations

### Disposal methods

|                                     |                                                                                                                 |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Waste from residues/unused products | Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation. |
| Contaminated packaging              | Do not reuse empty containers.                                                                                  |

## 14. Transport information

|             |               |
|-------------|---------------|
| <u>TDG</u>  | Not regulated |
| <u>DOT</u>  | Not regulated |
| <u>MEX</u>  | Not regulated |
| <u>IATA</u> | Not regulated |
| <u>IMDG</u> | Not regulated |

## 15. Regulatory information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Regulations

The Montreal Protocol on Substances that Deplete the Ozone Layer Not applicable

The Stockholm Convention on Persistent Organic Pollutants Not applicable

The Rotterdam Convention Not applicable

**International Inventories**

Contact supplier for inventory compliance status

**16. Other information**

|             |                         |                       |                           |                                           |
|-------------|-------------------------|-----------------------|---------------------------|-------------------------------------------|
| <b>NFPA</b> | <b>Health hazards</b> 2 | <b>Flammability</b> 0 | <b>Instability</b> 0      | <b>Physical and chemical properties -</b> |
| <b>HMIS</b> | <b>Health hazards</b> 2 | <b>Flammability</b> 0 | <b>Physical hazards</b> 0 | <b>Personal protection</b> X              |

**Key or legend to abbreviations and acronyms used in the safety data sheet****Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | Sk*  | Skin designation                 |

**Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
U.S. Environmental Protection Agency ChemView Database  
European Food Safety Authority (EFSA)  
Environmental Protection Agency  
Acute Exposure Guideline Level(s) (AEGL(s))  
U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
U.S. Environmental Protection Agency High Production Volume Chemicals  
Food Research Journal  
Hazardous Substance Database  
International Uniform Chemical Information Database (IUCLID)  
Japan GHS Classification  
Australia National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
NIOSH (National Institute for Occupational Safety and Health)  
National Library of Medicine's ChemID Plus (NLM CIP)  
National Library of Medicine's PubMed database (NLM PUBMED)  
U.S. National Toxicology Program (NTP)  
New Zealand's Chemical Classification and Information Database (CCID)  
Organization for Economic Co-operation and Development Environment, Health, and Safety Publications  
Organization for Economic Co-operation and Development High Production Volume Chemicals Program  
Organization for Economic Co-operation and Development Screening Information Data Set  
RTECS (Registry of Toxic Effects of Chemical Substances)  
World Health Organization

**Prepared By** Bio-Rad Laboratories, Environmental Health and Safety.

**Revision date** 21-Aug-2024

**Revision Note** Significant changes throughout SDS. Review all sections.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

**End of Safety Data Sheet**



# SAFETY DATA SHEET

This safety data sheet was created pursuant to the requirements of:  
Canada WHMIS 2015 which includes the amended Hazardous Products Act (HPA) and the Hazardous Products Regulation (HPR)

Revision date 21-Aug-2024

Revision Number 1.1

## 1. Identification

### Product identifier

Product Name C3 - 400 mIU/ml Calibrator, 2.2 ml

### Other means of identification

Catalog Number(s) 7441V

### Recommended use of the chemical and restrictions on use

Recommended use Restricted to professional users  
In vitro diagnostic

Restrictions on use No information available

### Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer Address

Bio-Rad  
3 boulevard Raymond Poincaré  
92430 Marnes-la-Coquette  
France  
e-mail: fds-msds.fr@bio-rad.com

#### Legal Entity / Contact Address

Bio-Rad Laboratories (Canada) Ltd.  
2403 Guenette  
Montreal, Quebec H4R 2E9  
Canada

Technical Service 1-800-361-1808  
CSD\_Techsupport@bio-rad.com

### Emergency telephone number

24 Hour Emergency Phone Number CHEMTREC Canada:1 (800) 424-9300

## 2. Hazard(s) identification

### Classification

Skin sensitization

Category 1

### Label elements

#### **Warning**

#### **Hazard statements**

May cause an allergic skin reaction



**Precautionary Statements - Prevention**

Avoid breathing dust/fume/gas/mist/vapors/spray  
Contaminated work clothing should not be allowed out of the workplace  
Wear protective gloves/protective clothing/eye protection/face protection

**Skin**

IF ON SKIN: Wash with plenty of water and soap  
If skin irritation or rash occurs: Get medical advice/attention  
Take off contaminated clothing and wash it before reuse

**Precautionary Statements - Disposal**

Dispose of contents/container in accordance with local, regional, national, and international regulations as applicable

**Other information**

Contains animal source material. (Cattle).  
Contains human source material and / or potentially infectious components

**3. Composition/information on ingredients****Substance**

Not applicable.

**Mixture**

The product contains no substances which at their given concentration, are considered to be hazardous to health

| Chemical name                                                                       | CAS No.    | Weight-%     | Hazardous Material<br>Information Review Act<br>registry number<br>(HMIRA registry #) | Date HMIRA filed and<br>date exemption granted<br>(if applicable) |
|-------------------------------------------------------------------------------------|------------|--------------|---------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with<br>2-methyl-3(2H)-isothiazolone | 55965-84-9 | 0.01 - 0.099 | -                                                                                     |                                                                   |

**4. First-aid measures****Description of first aid measures**

|                       |                                                                                                                                                                                                 |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General advice</b> | Show this safety data sheet to the doctor in attendance.                                                                                                                                        |
| <b>Inhalation</b>     | Remove to fresh air.                                                                                                                                                                            |
| <b>Eye contact</b>    | Contains human source material and / or potentially infectious components. Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids. Consult a physician. |
| <b>Skin contact</b>   | Wash with soap and water. May cause an allergic skin reaction. In the case of skin irritation or allergic reactions see a physician. Wash skin with soap and water.                             |
| <b>Ingestion</b>      | Call a physician. Contains human source material and / or potentially infectious components.                                                                                                    |

**Most important symptoms and effects, both acute and delayed**

**Symptoms** Itching. Rashes. Hives.

**Indication of any immediate medical attention and special treatment needed**

**Note to physicians** May cause sensitization in susceptible persons. Treat symptomatically. Contains human source material and / or potentially infectious components.

## 5. Fire-fighting measures

**Suitable Extinguishing Media** Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

**Unsuitable extinguishing media** No information available.

**Specific hazards arising from the chemical** Product is or contains a sensitizer. May cause sensitization by skin contact.

**Explosion data**

**Sensitivity to mechanical impact** None.

**Sensitivity to static discharge** None.

**Special protective equipment for fire-fighters** Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment.

## 6. Accidental release measures

**Personal precautions, protective equipment and emergency procedures**

**Personal precautions** Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak.

**Methods and material for containment and cleaning up**

**Methods for containment** Do not allow into any sewer, on the ground or into any body of water.

**Methods for cleaning up** Clean contaminated surface thoroughly. Use: Disinfectant.

## 7. Handling and storage

**Precautions for safe handling**

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice. Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. In case of insufficient ventilation, wear suitable respiratory equipment. Do not eat, drink or smoke when using this product. Take off contaminated clothing and wash before reuse.

**Conditions for safe storage, including any incompatibilities**

**Storage Conditions** Keep containers tightly closed in a dry, cool and well-ventilated place. Store locked up. Keep out of the reach of children. Store according to product and label instructions.

## 8. Exposure controls/personal protection

**Control parameters**

**Exposure Limits** This product, as supplied, does not contain any hazardous materials with occupational



exposure limits established by the region specific regulatory bodies.

#### Appropriate engineering controls

**Engineering controls**                      Showers  
                                                          Eyewash stations  
                                                          Ventilation systems.

#### Individual protection measures, such as personal protective equipment

**Eye/face protection**                      Wear safety glasses with side shields (or goggles).

**Hand protection**                              Wear suitable gloves.

**Skin and body protection**                      Wear suitable protective clothing.

**Respiratory protection**                      No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

**General hygiene considerations**                      Follow universal and standard precautions for handling potentially infectious materials.

## **9. Physical and chemical properties**

#### Information on basic physical and chemical properties

**Physical state**                                      Liquid  
**Appearance**                                      Liquid  
**Color**                                                  green  
**Odor**                                                  Characteristic  
**Odor threshold**                                      No information available

| <u>Property</u>                                | <u>Values</u>     | <u>Remarks • Method</u> |
|------------------------------------------------|-------------------|-------------------------|
| <b>pH</b>                                      |                   | None known              |
| <b>Melting point / freezing point</b>          | No data available | None known              |
| <b>Initial boiling point and boiling range</b> | No data available | None known              |
| <b>Flash point</b>                             | No data available | None known              |
| <b>Evaporation rate</b>                        | No data available | None known              |
| <b>Flammability</b>                            | No data available | None known              |
| <b>Flammability Limit in Air</b>               |                   | None known              |
| <b>Upper flammability or explosive limits</b>  | No data available |                         |
| <b>Lower flammability or explosive limits</b>  | No data available |                         |
| <b>Vapor pressure</b>                          | No data available | None known              |
| <b>Relative vapor density</b>                  | No data available | None known              |
| <b>Relative density</b>                        | No data available | None known              |
| <b>Water solubility</b>                        | Miscible in water |                         |
| <b>Solubility in other solvents</b>            | No data available | None known              |
| <b>Partition coefficient</b>                   | No data available | None known              |
| <b>Autoignition temperature</b>                | No data available | None known              |
| <b>Decomposition temperature</b>               |                   | None known              |
| <b>Kinematic viscosity</b>                     | No data available | None known              |
| <b>Dynamic viscosity</b>                       | No data available | None known              |

#### Other information

**Explosive properties**                              Not applicable.  
**Oxidizing properties**                              Not applicable.  
**Softening point**                                      Not applicable  
**Molecular weight**                                      Not applicable  
**VOC content**                                          Not applicable

**10. Stability and reactivity**

|                                    |                                           |
|------------------------------------|-------------------------------------------|
| Reactivity                         | No information available.                 |
| Chemical stability                 | Stable under normal conditions.           |
| Possibility of hazardous reactions | None under normal processing.             |
| Conditions to avoid                | None known based on information supplied. |
| Incompatible materials             | None known based on information supplied. |
| Hazardous decomposition products   | None known based on information supplied. |

**11. Toxicological information****Information on likely routes of exposure****Product Information**

|              |                                                                                                                                                                                                                           |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Inhalation   | Specific test data for the substance or mixture is not available.                                                                                                                                                         |
| Eye contact  | Specific test data for the substance or mixture is not available.                                                                                                                                                         |
| Skin contact | May cause sensitization by skin contact. Specific test data for the substance or mixture is not available Repeated or prolonged skin contact may cause allergic reactions with susceptible persons (based on components). |
| Ingestion    | Specific test data for the substance or mixture is not available.                                                                                                                                                         |

**Symptoms related to the physical, chemical and toxicological characteristics**

|          |                         |
|----------|-------------------------|
| Symptoms | Itching. Rashes. Hives. |
|----------|-------------------------|

**Acute toxicity**

**Numerical measures of toxicity**  
No information available

**Component Information**

| Chemical name                                                                                  | Oral LD50          | Dermal LD50              | Inhalation LC50 |
|------------------------------------------------------------------------------------------------|--------------------|--------------------------|-----------------|
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with 2-methyl-3(2H)-isothiazolone<br>55965-84-9 | = 53 mg/kg ( Rat ) | = 87.12 mg/kg ( Rabbit ) | -               |

**Delayed and immediate effects as well as chronic effects from short and long-term exposure**

|                                   |                                                                   |
|-----------------------------------|-------------------------------------------------------------------|
| Skin corrosion/irritation         | Based on available data, the classification criteria are not met. |
| Serious eye damage/eye irritation | Based on available data, the classification criteria are not met. |
| Respiratory or skin sensitization | May cause sensitization by skin contact.                          |
| Germ cell mutagenicity            | Based on available data, the classification criteria are not met. |
| Carcinogenicity                   | Based on available data, the classification criteria are not met. |
| Reproductive toxicity             | Based on available data, the classification criteria are not met. |

|                          |                                                                   |
|--------------------------|-------------------------------------------------------------------|
| STOT - single exposure   | Based on available data, the classification criteria are not met. |
| STOT - repeated exposure | Based on available data, the classification criteria are not met. |
| Aspiration hazard        | Based on available data, the classification criteria are not met. |

## 12. Ecological information

**Ecotoxicity** Harmful to aquatic life with long lasting effects.

**Persistence and degradability** No information available.

**Bioaccumulation** There is no data for this product.

### Component Information

| Chemical name                                                                                     | Partition coefficient |
|---------------------------------------------------------------------------------------------------|-----------------------|
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with<br>2-methyl-3(2H)-isothiazolone<br>55965-84-9 | 0.7                   |

**Other adverse effects** No information available.

## 13. Disposal considerations

### Disposal methods

**Waste from residues/unused products** Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation.

**Contaminated packaging** Do not reuse empty containers.

## 14. Transport information

**TDG** Not regulated

**DOT** Not regulated

**MEX** Not regulated

**IATA** Not regulated

**IMDG** Not regulated

## 15. Regulatory information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Regulations

**The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable

**The Stockholm Convention on Persistent Organic Pollutants** Not applicable

**The Rotterdam Convention** Not applicable

**International Inventories**

Contact supplier for inventory compliance status

**16. Other information**

|             |                         |                       |                           |                                           |
|-------------|-------------------------|-----------------------|---------------------------|-------------------------------------------|
| <b>NFPA</b> | <b>Health hazards</b> 2 | <b>Flammability</b> 0 | <b>Instability</b> 0      | <b>Physical and chemical properties -</b> |
| <b>HMIS</b> | <b>Health hazards</b> 2 | <b>Flammability</b> 0 | <b>Physical hazards</b> 0 | <b>Personal protection</b> X              |

**Key or legend to abbreviations and acronyms used in the safety data sheet****Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | Sk*  | Skin designation                 |

**Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
U.S. Environmental Protection Agency ChemView Database  
European Food Safety Authority (EFSA)  
Environmental Protection Agency  
Acute Exposure Guideline Level(s) (AEGL(s))  
U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
U.S. Environmental Protection Agency High Production Volume Chemicals  
Food Research Journal  
Hazardous Substance Database  
International Uniform Chemical Information Database (IUCLID)  
Japan GHS Classification  
Australia National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
NIOSH (National Institute for Occupational Safety and Health)  
National Library of Medicine's ChemID Plus (NLM CIP)  
National Library of Medicine's PubMed database (NLM PUBMED)  
U.S. National Toxicology Program (NTP)  
New Zealand's Chemical Classification and Information Database (CCID)  
Organization for Economic Co-operation and Development Environment, Health, and Safety Publications  
Organization for Economic Co-operation and Development High Production Volume Chemicals Program  
Organization for Economic Co-operation and Development Screening Information Data Set  
RTECS (Registry of Toxic Effects of Chemical Substances)  
World Health Organization

**Prepared By** Bio-Rad Laboratories, Environmental Health and Safety.

**Revision date** 21-Aug-2024

**Revision Note** Significant changes throughout SDS. Review all sections.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

**End of Safety Data Sheet**



# SAFETY DATA SHEET

This safety data sheet was created pursuant to the requirements of:  
Canada WHMIS 2015 which includes the amended Hazardous Products Act (HPA) and the Hazardous Products Regulation (HPR)

Revision date 21-Aug-2024

Revision Number 1.1

## 1. Identification

### Product identifier

Product Name C4 - 1000 mIU/ml Calibrator, 2.2 ml

### Other means of identification

Catalog Number(s) 7441F

### Recommended use of the chemical and restrictions on use

Recommended use Restricted to professional users  
In vitro diagnostic

Restrictions on use No information available

### Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer Address

Bio-Rad  
3 boulevard Raymond Poincaré  
92430 Marnes-la-Coquette  
France  
e-mail: fds-msds.fr@bio-rad.com

#### Legal Entity / Contact Address

Bio-Rad Laboratories (Canada) Ltd.  
2403 Guenette  
Montreal, Quebec H4R 2E9  
Canada

Technical Service 1-800-361-1808  
CSD\_Techsupport@bio-rad.com

### Emergency telephone number

24 Hour Emergency Phone Number CHEMTREC Canada:1 (800) 424-9300

## 2. Hazard(s) identification

### Classification

Skin sensitization

Category 1

### Label elements

#### **Warning**

#### **Hazard statements**

May cause an allergic skin reaction



**Precautionary Statements - Prevention**

Avoid breathing dust/fume/gas/mist/vapors/spray  
Contaminated work clothing should not be allowed out of the workplace  
Wear protective gloves/protective clothing/eye protection/face protection

**Skin**

IF ON SKIN: Wash with plenty of water and soap  
If skin irritation or rash occurs: Get medical advice/attention  
Take off contaminated clothing and wash it before reuse

**Precautionary Statements - Disposal**

Dispose of contents/container in accordance with local, regional, national, and international regulations as applicable

**Other information**

Contains animal source material. (Cattle).  
Contains human source material and / or potentially infectious components

**3. Composition/information on ingredients****Substance**

Not applicable.

**Mixture**

The product contains no substances which at their given concentration, are considered to be hazardous to health

| Chemical name                                                                       | CAS No.    | Weight-%     | Hazardous Material<br>Information Review Act<br>registry number<br>(HMIRA registry #) | Date HMIRA filed and<br>date exemption granted<br>(if applicable) |
|-------------------------------------------------------------------------------------|------------|--------------|---------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with<br>2-methyl-3(2H)-isothiazolone | 55965-84-9 | 0.01 - 0.099 | -                                                                                     |                                                                   |

**4. First-aid measures****Description of first aid measures**

|                       |                                                                                                                                                                                                 |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General advice</b> | Show this safety data sheet to the doctor in attendance.                                                                                                                                        |
| <b>Inhalation</b>     | Remove to fresh air.                                                                                                                                                                            |
| <b>Eye contact</b>    | Contains human source material and / or potentially infectious components. Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids. Consult a physician. |
| <b>Skin contact</b>   | Wash with soap and water. May cause an allergic skin reaction. In the case of skin irritation or allergic reactions see a physician. Wash skin with soap and water.                             |
| <b>Ingestion</b>      | Call a physician. Contains human source material and / or potentially infectious components.                                                                                                    |

**Most important symptoms and effects, both acute and delayed**

**Symptoms** Itching. Rashes. Hives.

**Indication of any immediate medical attention and special treatment needed**

**Note to physicians** May cause sensitization in susceptible persons. Treat symptomatically. Contains human source material and / or potentially infectious components.

## 5. Fire-fighting measures

**Suitable Extinguishing Media** Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

**Unsuitable extinguishing media** No information available.

**Specific hazards arising from the chemical** Product is or contains a sensitizer. May cause sensitization by skin contact.

**Explosion data**

**Sensitivity to mechanical impact** None.

**Sensitivity to static discharge** None.

**Special protective equipment for fire-fighters** Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment.

## 6. Accidental release measures

**Personal precautions, protective equipment and emergency procedures**

**Personal precautions** Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak.

**Methods and material for containment and cleaning up**

**Methods for containment** Do not allow into any sewer, on the ground or into any body of water.

**Methods for cleaning up** Clean contaminated surface thoroughly. Use: Disinfectant.

## 7. Handling and storage

**Precautions for safe handling**

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice. Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. In case of insufficient ventilation, wear suitable respiratory equipment. Do not eat, drink or smoke when using this product. Take off contaminated clothing and wash before reuse.

**Conditions for safe storage, including any incompatibilities**

**Storage Conditions** Keep containers tightly closed in a dry, cool and well-ventilated place. Store locked up. Keep out of the reach of children. Store according to product and label instructions.

## 8. Exposure controls/personal protection

**Control parameters**

**Exposure Limits** This product, as supplied, does not contain any hazardous materials with occupational

exposure limits established by the region specific regulatory bodies.

### Appropriate engineering controls

|                             |                      |
|-----------------------------|----------------------|
| <b>Engineering controls</b> | Showers              |
|                             | Eyewash stations     |
|                             | Ventilation systems. |

### Individual protection measures, such as personal protective equipment

|                                       |                                                                                                                                                                          |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye/face protection</b>            | Wear safety glasses with side shields (or goggles).                                                                                                                      |
| <b>Hand protection</b>                | Wear suitable gloves.                                                                                                                                                    |
| <b>Skin and body protection</b>       | Wear suitable protective clothing.                                                                                                                                       |
| <b>Respiratory protection</b>         | No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required. |
| <b>General hygiene considerations</b> | Follow universal and standard precautions for handling potentially infectious materials.                                                                                 |

## 9. Physical and chemical properties

### Information on basic physical and chemical properties

|                       |                          |
|-----------------------|--------------------------|
| <b>Physical state</b> | Liquid                   |
| <b>Appearance</b>     | Liquid                   |
| <b>Color</b>          | dark green               |
| <b>Odor</b>           | Characteristic           |
| <b>Odor threshold</b> | No information available |

| <u>Property</u>                         | <u>Values</u>     | <u>Remarks • Method</u> |
|-----------------------------------------|-------------------|-------------------------|
| pH                                      |                   | None known              |
| Melting point / freezing point          | No data available | None known              |
| Initial boiling point and boiling range | No data available | None known              |
| Flash point                             | No data available | None known              |
| Evaporation rate                        | No data available | None known              |
| Flammability                            | No data available | None known              |
| Flammability Limit in Air               |                   | None known              |
| Upper flammability or explosive limits  | No data available |                         |
| Lower flammability or explosive limits  | No data available |                         |
| Vapor pressure                          | No data available | None known              |
| Relative vapor density                  | No data available | None known              |
| Relative density                        | No data available | None known              |
| Water solubility                        | Miscible in water |                         |
| Solubility in other solvents            | No data available | None known              |
| Partition coefficient                   | No data available | None known              |
| Autoignition temperature                | No data available | None known              |
| Decomposition temperature               |                   | None known              |
| Kinematic viscosity                     | No data available | None known              |
| Dynamic viscosity                       | No data available | None known              |

## Other information

|                             |                 |
|-----------------------------|-----------------|
| <b>Explosive properties</b> | Not applicable. |
| <b>Oxidizing properties</b> | Not applicable. |
| <b>Softening point</b>      | Not applicable  |
| <b>Molecular weight</b>     | Not applicable  |
| <b>VOC content</b>          | Not applicable  |



**10. Stability and reactivity**

|                                    |                                           |
|------------------------------------|-------------------------------------------|
| Reactivity                         | No information available.                 |
| Chemical stability                 | Stable under normal conditions.           |
| Possibility of hazardous reactions | None under normal processing.             |
| Conditions to avoid                | None known based on information supplied. |
| Incompatible materials             | None known based on information supplied. |
| Hazardous decomposition products   | None known based on information supplied. |

**11. Toxicological information****Information on likely routes of exposure****Product Information**

|              |                                                                                                                                                                                                                           |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Inhalation   | Specific test data for the substance or mixture is not available.                                                                                                                                                         |
| Eye contact  | Specific test data for the substance or mixture is not available.                                                                                                                                                         |
| Skin contact | May cause sensitization by skin contact. Specific test data for the substance or mixture is not available Repeated or prolonged skin contact may cause allergic reactions with susceptible persons (based on components). |
| Ingestion    | Specific test data for the substance or mixture is not available.                                                                                                                                                         |

**Symptoms related to the physical, chemical and toxicological characteristics**

|          |                         |
|----------|-------------------------|
| Symptoms | Itching. Rashes. Hives. |
|----------|-------------------------|

**Acute toxicity**

**Numerical measures of toxicity**  
No information available

**Component Information**

| Chemical name                                                                                  | Oral LD50          | Dermal LD50              | Inhalation LC50 |
|------------------------------------------------------------------------------------------------|--------------------|--------------------------|-----------------|
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with 2-methyl-3(2H)-isothiazolone<br>55965-84-9 | = 53 mg/kg ( Rat ) | = 87.12 mg/kg ( Rabbit ) | -               |

**Delayed and immediate effects as well as chronic effects from short and long-term exposure**

|                                   |                                                                   |
|-----------------------------------|-------------------------------------------------------------------|
| Skin corrosion/irritation         | Based on available data, the classification criteria are not met. |
| Serious eye damage/eye irritation | Based on available data, the classification criteria are not met. |
| Respiratory or skin sensitization | May cause sensitization by skin contact.                          |
| Germ cell mutagenicity            | Based on available data, the classification criteria are not met. |
| Carcinogenicity                   | Based on available data, the classification criteria are not met. |
| Reproductive toxicity             | Based on available data, the classification criteria are not met. |

|                          |                                                                   |
|--------------------------|-------------------------------------------------------------------|
| STOT - single exposure   | Based on available data, the classification criteria are not met. |
| STOT - repeated exposure | Based on available data, the classification criteria are not met. |
| Aspiration hazard        | Based on available data, the classification criteria are not met. |

## 12. Ecological information

|                               |                                                    |
|-------------------------------|----------------------------------------------------|
| Ecotoxicity                   | Harmful to aquatic life with long lasting effects. |
| Persistence and degradability | No information available.                          |
| Bioaccumulation               | There is no data for this product.                 |

### Component Information

| Chemical name                                                                                     | Partition coefficient |
|---------------------------------------------------------------------------------------------------|-----------------------|
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with<br>2-methyl-3(2H)-isothiazolone<br>55965-84-9 | 0.7                   |

|                       |                           |
|-----------------------|---------------------------|
| Other adverse effects | No information available. |
|-----------------------|---------------------------|

## 13. Disposal considerations

### Disposal methods

|                                     |                                                                                                                 |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Waste from residues/unused products | Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation. |
| Contaminated packaging              | Do not reuse empty containers.                                                                                  |

## 14. Transport information

|             |               |
|-------------|---------------|
| <u>TDG</u>  | Not regulated |
| <u>DOT</u>  | Not regulated |
| <u>MEX</u>  | Not regulated |
| <u>IATA</u> | Not regulated |
| <u>IMDG</u> | Not regulated |

## 15. Regulatory information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Regulations

The Montreal Protocol on Substances that Deplete the Ozone Layer Not applicable

The Stockholm Convention on Persistent Organic Pollutants Not applicable

The Rotterdam Convention Not applicable

**International Inventories**

Contact supplier for inventory compliance status

**16. Other information**

|             |                         |                       |                           |                                           |
|-------------|-------------------------|-----------------------|---------------------------|-------------------------------------------|
| <b>NFPA</b> | <b>Health hazards</b> 2 | <b>Flammability</b> 0 | <b>Instability</b> 0      | <b>Physical and chemical properties -</b> |
| <b>HMIS</b> | <b>Health hazards</b> 2 | <b>Flammability</b> 0 | <b>Physical hazards</b> 0 | <b>Personal protection</b> X              |

**Key or legend to abbreviations and acronyms used in the safety data sheet****Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | Sk*  | Skin designation                 |

**Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
 U.S. Environmental Protection Agency ChemView Database  
 European Food Safety Authority (EFSA)  
 Environmental Protection Agency  
 Acute Exposure Guideline Level(s) (AEGL(s))  
 U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
 U.S. Environmental Protection Agency High Production Volume Chemicals  
 Food Research Journal  
 Hazardous Substance Database  
 International Uniform Chemical Information Database (IUCLID)  
 Japan GHS Classification  
 Australia National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
 NIOSH (National Institute for Occupational Safety and Health)  
 National Library of Medicine's ChemID Plus (NLM CIP)  
 National Library of Medicine's PubMed database (NLM PUBMED)  
 U.S. National Toxicology Program (NTP)  
 New Zealand's Chemical Classification and Information Database (CCID)  
 Organization for Economic Co-operation and Development Environment, Health, and Safety Publications  
 Organization for Economic Co-operation and Development High Production Volume Chemicals Program  
 Organization for Economic Co-operation and Development Screening Information Data Set  
 RTECS (Registry of Toxic Effects of Chemical Substances)  
 World Health Organization

**Prepared By** Bio-Rad Laboratories, Environmental Health and Safety.

**Revision date** 21-Aug-2024

**Revision Note** Significant changes throughout SDS. Review all sections.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

**End of Safety Data Sheet**



# SAFETY DATA SHEET

This safety data sheet was created pursuant to the requirements of:  
Canada WHMIS 2015 which includes the amended Hazardous Products Act (HPA) and the Hazardous Products Regulation (HPR)

Revision date 14-Aug-2024

Revision Number 1.1

## 1. Identification

### Product identifier

Product Name R6 - Specimen Diluent, 27 ml

### Other means of identification

Catalog Number(s) 7441B

### Recommended use of the chemical and restrictions on use

Recommended use Restricted to professional users  
In vitro diagnostic

Restrictions on use No information available

### Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer Address

Bio-Rad  
3 boulevard Raymond Poincaré  
92430 Marnes-la-Coquette  
France  
e-mail: fds-msds.fr@bio-rad.com

#### Legal Entity / Contact Address

Bio-Rad Laboratories (Canada) Ltd.  
2403 Guenette  
Montreal, Quebec H4R 2E9  
Canada

Technical Service 1-800-361-1808  
CSD\_Techsupport@bio-rad.com

### Emergency telephone number

24 Hour Emergency Phone Number CHEMTREC Canada:1 (800) 424-9300

## 2. Hazard(s) identification

### Classification

|                    |            |
|--------------------|------------|
| Skin sensitization | Category 1 |
|--------------------|------------|

### Label elements

#### **Warning**

#### **Hazard statements**

May cause an allergic skin reaction



**Precautionary Statements - Prevention**

Avoid breathing dust/fume/gas/mist/vapors/spray  
Contaminated work clothing should not be allowed out of the workplace  
Wear protective gloves/protective clothing/eye protection/face protection

**Skin**

IF ON SKIN: Wash with plenty of water and soap  
If skin irritation or rash occurs: Get medical advice/attention  
Take off contaminated clothing and wash it before reuse

**Precautionary Statements - Disposal**

Dispose of contents/container in accordance with local, regional, national, and international regulations as applicable

**Other information**

Contains animal source material. (Cattle).

**3. Composition/information on ingredients****Substance**

Not applicable.

**Mixture**

The product contains no substances which at their given concentration, are considered to be hazardous to health

| Chemical name                                                                       | CAS No.    | Weight-%     | Hazardous Material<br>Information Review Act<br>registry number<br>(HMIRA registry #) | Date HMIRA filed and<br>date exemption granted<br>(if applicable) |
|-------------------------------------------------------------------------------------|------------|--------------|---------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with<br>2-methyl-3(2H)-isothiazolone | 55965-84-9 | 0.001 - 0.01 | -                                                                                     |                                                                   |

**4. First-aid measures****Description of first aid measures**

|                       |                                                                                                                                         |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>General advice</b> | Show this safety data sheet to the doctor in attendance.                                                                                |
| <b>Inhalation</b>     | Remove to fresh air.                                                                                                                    |
| <b>Eye contact</b>    | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids.<br>Consult a physician.                 |
| <b>Skin contact</b>   | Wash with soap and water. May cause an allergic skin reaction. In the case of skin irritation<br>or allergic reactions see a physician. |
| <b>Ingestion</b>      | Rinse mouth thoroughly with water.                                                                                                      |

**Most important symptoms and effects, both acute and delayed**

|                 |                         |
|-----------------|-------------------------|
| <b>Symptoms</b> | Itching. Rashes. Hives. |
|-----------------|-------------------------|

**Indication of any immediate medical attention and special treatment needed**

**Note to physicians** May cause sensitization in susceptible persons. Treat symptomatically.

## 5. Fire-fighting measures

|                                                       |                                                                                                                                    |
|-------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>                   | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.                            |
| <b>Unsuitable extinguishing media</b>                 | No information available.                                                                                                          |
| <b>Specific hazards arising from the chemical</b>     | Product is or contains a sensitizer. May cause sensitization by skin contact.                                                      |
| <b>Explosion data</b>                                 |                                                                                                                                    |
| <b>Sensitivity to mechanical impact</b>               | None.                                                                                                                              |
| <b>Sensitivity to static discharge</b>                | None.                                                                                                                              |
| <b>Special protective equipment for fire-fighters</b> | Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment. |

## 6. Accidental release measures

### Personal precautions, protective equipment and emergency procedures

|                             |                                                                                                                                                                                                          |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal precautions</b> | Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Methods and material for containment and cleaning up

|                                |                                                       |
|--------------------------------|-------------------------------------------------------|
| <b>Methods for containment</b> | Prevent further leakage or spillage if safe to do so. |
| <b>Methods for cleaning up</b> | Pick up and transfer to properly labeled containers.  |

## 7. Handling and storage

### Precautions for safe handling

|                                |                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Advice on safe handling</b> | Handle in accordance with good industrial hygiene and safety practice. Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. In case of insufficient ventilation, wear suitable respiratory equipment. Do not eat, drink or smoke when using this product. Take off contaminated clothing and wash before reuse. |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Conditions for safe storage, including any incompatibilities

|                           |                                                                                                                                                                                 |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Storage Conditions</b> | Keep containers tightly closed in a dry, cool and well-ventilated place. Store locked up. Keep out of the reach of children. Store according to product and label instructions. |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 8. Exposure controls/personal protection

### Control parameters

|                        |                                                                                                                                                             |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Exposure Limits</b> | This product, as supplied, does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies. |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Appropriate engineering controls

**Engineering controls**

Showers  
Eyewash stations  
Ventilation systems.

#### **Individual protection measures, such as personal protective equipment**

**Eye/face protection** Wear safety glasses with side shields (or goggles).

**Hand protection** Wear suitable gloves.

**Skin and body protection** Wear suitable protective clothing.

**Respiratory protection** No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

**General hygiene considerations** Handle in accordance with good industrial hygiene and safety practice.

## **9. Physical and chemical properties**

### **Information on basic physical and chemical properties**

**Physical state** Liquid  
**Appearance** Liquid  
**Color** violet  
**Odor** Low  
**Odor threshold** No information available

| <b><u>Property</u></b>                         | <b><u>Values</u></b> | <b><u>Remarks • Method</u></b> |
|------------------------------------------------|----------------------|--------------------------------|
| <b>pH</b>                                      | 7.5                  | None known                     |
| <b>Melting point / freezing point</b>          | No data available    | None known                     |
| <b>Initial boiling point and boiling range</b> | No data available    | None known                     |
| <b>Flash point</b>                             | No data available    | None known                     |
| <b>Evaporation rate</b>                        | No data available    | None known                     |
| <b>Flammability</b>                            | No data available    | None known                     |
| <b>Flammability Limit in Air</b>               |                      | None known                     |
| <b>Upper flammability or explosive limits</b>  | No data available    |                                |
| <b>Lower flammability or explosive limits</b>  | No data available    |                                |
| <b>Vapor pressure</b>                          | No data available    | None known                     |
| <b>Relative vapor density</b>                  | No data available    | None known                     |
| <b>Relative density</b>                        | No data available    | None known                     |
| <b>Water solubility</b>                        | Immiscible in water  |                                |
| <b>Solubility in other solvents</b>            | No data available    | None known                     |
| <b>Partition coefficient</b>                   | No data available    | None known                     |
| <b>Autoignition temperature</b>                | No data available    | None known                     |
| <b>Decomposition temperature</b>               |                      | None known                     |
| <b>Kinematic viscosity</b>                     | No data available    | None known                     |
| <b>Dynamic viscosity</b>                       | No data available    | None known                     |

### **Other information**

**Explosive properties** Not applicable.  
**Oxidizing properties** Not applicable.  
**Softening point** Not applicable  
**Molecular weight** Not applicable  
**VOC content** Not applicable

## **10. Stability and reactivity**

**Reactivity** No information available.

|                                    |                                           |
|------------------------------------|-------------------------------------------|
| Chemical stability                 | Stable under normal conditions.           |
| Possibility of hazardous reactions | None under normal processing.             |
| Conditions to avoid                | None known based on information supplied. |
| Incompatible materials             | None known based on information supplied. |
| Hazardous decomposition products   | None known based on information supplied. |

## 11. Toxicological information

### Information on likely routes of exposure

#### Product Information

|              |                                                                                                                                                                                                                            |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Inhalation   | Specific test data for the substance or mixture is not available.                                                                                                                                                          |
| Eye contact  | Specific test data for the substance or mixture is not available.                                                                                                                                                          |
| Skin contact | May cause sensitization by skin contact. Specific test data for the substance or mixture is not available. Repeated or prolonged skin contact may cause allergic reactions with susceptible persons (based on components). |
| Ingestion    | Specific test data for the substance or mixture is not available.                                                                                                                                                          |

### Symptoms related to the physical, chemical and toxicological characteristics

|          |                         |
|----------|-------------------------|
| Symptoms | Itching. Rashes. Hives. |
|----------|-------------------------|

### Acute toxicity

#### Numerical measures of toxicity

No information available

The following values are calculated based on chapter 3.1 of the GHS document

ATEmix (oral) 3,700,000.00 mg/kg

#### Component Information

| Chemical name                                                                                  | Oral LD50          | Dermal LD50              | Inhalation LC50 |
|------------------------------------------------------------------------------------------------|--------------------|--------------------------|-----------------|
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with 2-methyl-3(2H)-isothiazolone<br>55965-84-9 | = 53 mg/kg ( Rat ) | = 87.12 mg/kg ( Rabbit ) | -               |

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                                   |                                                                   |
|-----------------------------------|-------------------------------------------------------------------|
| Skin corrosion/irritation         | Based on available data, the classification criteria are not met. |
| Serious eye damage/eye irritation | Based on available data, the classification criteria are not met. |
| Respiratory or skin sensitization | May cause sensitization by skin contact.                          |
| Germ cell mutagenicity            | Based on available data, the classification criteria are not met. |
| Carcinogenicity                   | Based on available data, the classification criteria are not met. |
| Reproductive toxicity             | Based on available data, the classification criteria are not met. |
| STOT - single exposure            | Based on available data, the classification criteria are not met. |



**STOT - repeated exposure** Based on available data, the classification criteria are not met.

**Aspiration hazard** Based on available data, the classification criteria are not met.

## 12. Ecological information

**Ecotoxicity** Harmful to aquatic life with long lasting effects.

**Persistence and degradability** No information available.

**Bioaccumulation** There is no data for this product.

### Component Information

| Chemical name                                                                                     | Partition coefficient |
|---------------------------------------------------------------------------------------------------|-----------------------|
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with<br>2-methyl-3(2H)-isothiazolone<br>55965-84-9 | 0.7                   |

**Other adverse effects** No information available.

## 13. Disposal considerations

### Disposal methods

**Waste from residues/unused products** Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation.

**Contaminated packaging** Do not reuse empty containers.

## 14. Transport information

**TDG** Not regulated

**DOT** Not regulated

**MEX** Not regulated

**IATA** Not regulated

**IMDG** Not regulated

## 15. Regulatory information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Regulations

**The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable

**The Stockholm Convention on Persistent Organic Pollutants** Not applicable

**The Rotterdam Convention** Not applicable

#### International Inventories

Contact supplier for inventory compliance status

**16. Other information**

|             |                         |                       |                           |                                           |
|-------------|-------------------------|-----------------------|---------------------------|-------------------------------------------|
| <b>NFPA</b> | <b>Health hazards</b> 2 | <b>Flammability</b> 0 | <b>Instability</b> 0      | <b>Physical and chemical properties</b> - |
| <b>HMIS</b> | <b>Health hazards</b> 2 | <b>Flammability</b> 0 | <b>Physical hazards</b> 0 | <b>Personal protection</b> X              |

**Key or legend to abbreviations and acronyms used in the safety data sheet****Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | Sk*  | Skin designation                 |

**Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
 U.S. Environmental Protection Agency ChemView Database  
 European Food Safety Authority (EFSA)  
 Environmental Protection Agency  
 Acute Exposure Guideline Level(s) (AEGL(s))  
 U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
 U.S. Environmental Protection Agency High Production Volume Chemicals  
 Food Research Journal  
 Hazardous Substance Database  
 International Uniform Chemical Information Database (IUCLID)  
 Japan GHS Classification  
 Australia National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
 NIOSH (National Institute for Occupational Safety and Health)  
 National Library of Medicine's ChemID Plus (NLM CIP)  
 National Library of Medicine's PubMed database (NLM PUBMED)  
 U.S. National Toxicology Program (NTP)  
 New Zealand's Chemical Classification and Information Database (CCID)  
 Organization for Economic Co-operation and Development Environment, Health, and Safety Publications  
 Organization for Economic Co-operation and Development High Production Volume Chemicals Program  
 Organization for Economic Co-operation and Development Screening Information Data Set  
 RTECS (Registry of Toxic Effects of Chemical Substances)  
 World Health Organization

**Prepared By** Bio-Rad Laboratories, Environmental Health and Safety.

**Revision date** 14-Aug-2024

**Revision Note** Significant changes throughout SDS. Review all sections.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

**End of Safety Data Sheet**



# SAFETY DATA SHEET

This safety data sheet was created pursuant to the requirements of:  
Canada WHMIS 2015 which includes the amended Hazardous Products Act (HPA) and the Hazardous Products Regulation (HPR)

Revision date 14-Aug-2024

Revision Number 1.1

## 1. Identification

### Product identifier

Product Name R7a - Conjugate (11X), 2.5 ml

### Other means of identification

Catalog Number(s) 7441G

### Recommended use of the chemical and restrictions on use

Recommended use Restricted to professional users  
In vitro diagnostic

Restrictions on use No information available

### Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer Address

Bio-Rad  
3 boulevard Raymond Poincaré  
92430 Marnes-la-Coquette  
France  
e-mail: fds-msds.fr@bio-rad.com

#### Legal Entity / Contact Address

Bio-Rad Laboratories (Canada) Ltd.  
2403 Guenette  
Montreal, Quebec H4R 2E9  
Canada

Technical Service 1-800-361-1808  
CSD\_Techsupport@bio-rad.com

### Emergency telephone number

24 Hour Emergency Phone Number CHEMTREC Canada:1 (800) 424-9300

## 2. Hazard(s) identification

### Classification

|                    |            |
|--------------------|------------|
| Skin sensitization | Category 1 |
|--------------------|------------|

### Label elements

#### **Warning**

#### **Hazard statements**

May cause an allergic skin reaction



**Precautionary Statements - Prevention**

Avoid breathing dust/fume/gas/mist/vapors/spray  
Contaminated work clothing should not be allowed out of the workplace  
Wear protective gloves/protective clothing/eye protection/face protection

**Skin**

IF ON SKIN: Wash with plenty of water and soap  
If skin irritation or rash occurs: Get medical advice/attention  
Take off contaminated clothing and wash it before reuse

**Precautionary Statements - Disposal**

Dispose of contents/container in accordance with local, regional, national, and international regulations as applicable

**Other information**

Contains animal source material. (Goat). (Cattle).

**3. Composition/information on ingredients****Substance**

Not applicable.

**Mixture**

| Chemical name                                                                       | CAS No.    | Weight-%     | Hazardous Material<br>Information Review Act<br>registry number<br>(HMIRA registry #) | Date HMIRA filed and<br>date exemption granted<br>(if applicable) |
|-------------------------------------------------------------------------------------|------------|--------------|---------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| 1,2,3-Propanetriol                                                                  | 56-81-5    | 20 - 35      | -                                                                                     |                                                                   |
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with<br>2-methyl-3(2H)-isothiazolone | 55965-84-9 | 0.01 - 0.099 | -                                                                                     |                                                                   |

**4. First-aid measures****Description of first aid measures**

|                       |                                                                                                                                         |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>General advice</b> | Show this safety data sheet to the doctor in attendance.                                                                                |
| <b>Inhalation</b>     | Remove to fresh air.                                                                                                                    |
| <b>Eye contact</b>    | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids.<br>Consult a physician.                 |
| <b>Skin contact</b>   | Wash with soap and water. May cause an allergic skin reaction. In the case of skin irritation<br>or allergic reactions see a physician. |
| <b>Ingestion</b>      | Rinse mouth thoroughly with water.                                                                                                      |

**Most important symptoms and effects, both acute and delayed**

|                 |                         |
|-----------------|-------------------------|
| <b>Symptoms</b> | Itching. Rashes. Hives. |
|-----------------|-------------------------|

**Indication of any immediate medical attention and special treatment needed**

**Note to physicians**

May cause sensitization in susceptible persons. Treat symptomatically.

**5. Fire-fighting measures****Suitable Extinguishing Media**

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

**Unsuitable extinguishing media**

No information available.

**Specific hazards arising from the chemical**

Product is or contains a sensitizer. May cause sensitization by skin contact.

**Explosion data**

**Sensitivity to mechanical impact** None.

**Sensitivity to static discharge** None.

**Special protective equipment for fire-fighters**

Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment.

**6. Accidental release measures****Personal precautions, protective equipment and emergency procedures****Personal precautions**

Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak.

**Methods and material for containment and cleaning up****Methods for containment**

Prevent further leakage or spillage if safe to do so.

**Methods for cleaning up**

Pick up and transfer to properly labeled containers.

**7. Handling and storage****Precautions for safe handling****Advice on safe handling**

Handle in accordance with good industrial hygiene and safety practice. Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. In case of insufficient ventilation, wear suitable respiratory equipment. Do not eat, drink or smoke when using this product. Take off contaminated clothing and wash before reuse.

**Conditions for safe storage, including any incompatibilities****Storage Conditions**

Keep containers tightly closed in a dry, cool and well-ventilated place. Store locked up. Keep out of the reach of children. Store according to product and label instructions.

**8. Exposure controls/personal protection****Control parameters****Exposure Limits**

| Chemical name                 | Alberta                   | British Columbia                                      | Ontario | Quebec                    |
|-------------------------------|---------------------------|-------------------------------------------------------|---------|---------------------------|
| 1,2,3-Propanetriol<br>56-81-5 | TWA: 10 mg/m <sup>3</sup> | TWA: 10 mg/m <sup>3</sup><br>TWA: 3 mg/m <sup>3</sup> |         | TWA: 10 mg/m <sup>3</sup> |

**Appropriate engineering controls**

**Engineering controls**                      Showers  
                                                          Eyewash stations  
                                                          Ventilation systems.

**Individual protection measures, such as personal protective equipment**

**Eye/face protection**                      Wear safety glasses with side shields (or goggles).

**Hand protection**                              Wear suitable gloves.

**Skin and body protection**                      Wear suitable protective clothing.

**Respiratory protection**                      No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

**General hygiene considerations**                      Handle in accordance with good industrial hygiene and safety practice.

**9. Physical and chemical properties****Information on basic physical and chemical properties**

**Physical state**                                      Liquid  
**Appearance**                                      viscous  
**Color**                                                  green  
**Odor**                                                  Low  
**Odor threshold**                                      No information available

| <b><u>Property</u></b>                         | <b><u>Values</u></b> | <b><u>Remarks • Method</u></b> |
|------------------------------------------------|----------------------|--------------------------------|
| <b>pH</b>                                      | 7.1                  | None known                     |
| <b>Melting point / freezing point</b>          | No data available    | None known                     |
| <b>Initial boiling point and boiling range</b> | No data available    | None known                     |
| <b>Flash point</b>                             | No data available    | None known                     |
| <b>Evaporation rate</b>                        | No data available    | None known                     |
| <b>Flammability</b>                            | No data available    | None known                     |
| <b>Flammability Limit in Air</b>               |                      | None known                     |
| <b>Upper flammability or explosive limits</b>  | No data available    |                                |
| <b>Lower flammability or explosive limits</b>  | No data available    |                                |
| <b>Vapor pressure</b>                          | No data available    | None known                     |
| <b>Relative vapor density</b>                  | No data available    | None known                     |
| <b>Relative density</b>                        | No data available    | None known                     |
| <b>Water solubility</b>                        | Miscible in water    |                                |
| <b>Solubility in other solvents</b>            | No data available    | None known                     |
| <b>Partition coefficient</b>                   | No data available    | None known                     |
| <b>Autoignition temperature</b>                | No data available    | None known                     |
| <b>Decomposition temperature</b>               |                      | None known                     |
| <b>Kinematic viscosity</b>                     | No data available    | None known                     |
| <b>Dynamic viscosity</b>                       | No data available    | None known                     |

**Other information**

**Explosive properties**                              Not applicable.  
**Oxidizing properties**                              Not applicable.  
**Softening point**                                      Not applicable  
**Molecular weight**                                      Not applicable  
**VOC content**                                              Not applicable

**10. Stability and reactivity**

|                                    |                                           |
|------------------------------------|-------------------------------------------|
| Reactivity                         | No information available.                 |
| Chemical stability                 | Stable under normal conditions.           |
| Possibility of hazardous reactions | None under normal processing.             |
| Conditions to avoid                | None known based on information supplied. |
| Incompatible materials             | None known based on information supplied. |
| Hazardous decomposition products   | None known based on information supplied. |

## 11. Toxicological information

### Information on likely routes of exposure

#### Product Information

|              |                                                                                                                                                                                                                           |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Inhalation   | Specific test data for the substance or mixture is not available.                                                                                                                                                         |
| Eye contact  | Specific test data for the substance or mixture is not available.                                                                                                                                                         |
| Skin contact | May cause sensitization by skin contact. Specific test data for the substance or mixture is not available Repeated or prolonged skin contact may cause allergic reactions with susceptible persons (based on components). |
| Ingestion    | Specific test data for the substance or mixture is not available.                                                                                                                                                         |

### Symptoms related to the physical, chemical and toxicological characteristics

|          |                         |
|----------|-------------------------|
| Symptoms | Itching. Rashes. Hives. |
|----------|-------------------------|

### Acute toxicity

#### Numerical measures of toxicity

No information available

#### Component Information

| Chemical name                                                                                     | Oral LD50             | Dermal LD50              | Inhalation LC50         |
|---------------------------------------------------------------------------------------------------|-----------------------|--------------------------|-------------------------|
| 1,2,3-Propanetriol<br>56-81-5                                                                     | = 12600 mg/kg ( Rat ) | > 10 g/kg ( Rabbit )     | > 2.75 mg/L ( Rat ) 4 h |
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with<br>2-methyl-3(2H)-isothiazolone<br>55965-84-9 | = 53 mg/kg ( Rat )    | = 87.12 mg/kg ( Rabbit ) | -                       |

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                                   |                                                                   |
|-----------------------------------|-------------------------------------------------------------------|
| Skin corrosion/irritation         | Based on available data, the classification criteria are not met. |
| Serious eye damage/eye irritation | Based on available data, the classification criteria are not met. |
| Respiratory or skin sensitization | May cause sensitization by skin contact.                          |
| Germ cell mutagenicity            | Based on available data, the classification criteria are not met. |
| Carcinogenicity                   | Based on available data, the classification criteria are not met. |
| Reproductive toxicity             | Based on available data, the classification criteria are not met. |

|                                 |                                                                   |
|---------------------------------|-------------------------------------------------------------------|
| <b>STOT - single exposure</b>   | Based on available data, the classification criteria are not met. |
| <b>STOT - repeated exposure</b> | Based on available data, the classification criteria are not met. |
| <b>Target organ effects</b>     | Kidney, Respiratory system, Eyes, Skin.                           |
| <b>Aspiration hazard</b>        | Based on available data, the classification criteria are not met. |

## 12. Ecological information

**Ecotoxicity** Harmful to aquatic life with long lasting effects.

| Chemical name                 | Algae/aquatic plants | Fish                                            | Toxicity to microorganisms | Crustacea |
|-------------------------------|----------------------|-------------------------------------------------|----------------------------|-----------|
| 1,2,3-Propanetriol<br>56-81-5 | -                    | LC50: 51 - 57mL/L (96h,<br>Oncorhynchus mykiss) | -                          | -         |

**Persistence and degradability** No information available.

**Bioaccumulation** There is no data for this product.

### Component Information

| Chemical name                                                                                     | Partition coefficient |
|---------------------------------------------------------------------------------------------------|-----------------------|
| 1,2,3-Propanetriol<br>56-81-5                                                                     | -1.75                 |
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with<br>2-methyl-3(2H)-isothiazolone<br>55965-84-9 | 0.7                   |

**Other adverse effects** No information available.

## 13. Disposal considerations

### Disposal methods

**Waste from residues/unused products** Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation.

**Contaminated packaging** Do not reuse empty containers.

## 14. Transport information

**TDG** Not regulated

**DOT** Not regulated

**MEX** Not regulated

**IATA** Not regulated

**IMDG** Not regulated

## 15. Regulatory information

**Safety, health and environmental regulations/legislation specific for the substance or mixture**



**International Regulations****The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable**The Stockholm Convention on Persistent Organic Pollutants** Not applicable**The Rotterdam Convention** Not applicable**International Inventories**

Contact supplier for inventory compliance status

**16. Other information**

|             |                         |                       |                           |                                           |
|-------------|-------------------------|-----------------------|---------------------------|-------------------------------------------|
| <b>NFPA</b> | <b>Health hazards</b> 2 | <b>Flammability</b> 0 | <b>Instability</b> 0      | <b>Physical and chemical properties -</b> |
| <b>HMIS</b> | <b>Health hazards</b> 2 | <b>Flammability</b> 0 | <b>Physical hazards</b> 0 | <b>Personal protection</b> X              |

**Key or legend to abbreviations and acronyms used in the safety data sheet****Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | Sk*  | Skin designation                 |

**Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
U.S. Environmental Protection Agency ChemView Database  
European Food Safety Authority (EFSA)  
Environmental Protection Agency  
Acute Exposure Guideline Level(s) (AEGL(s))  
U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
U.S. Environmental Protection Agency High Production Volume Chemicals  
Food Research Journal  
Hazardous Substance Database  
International Uniform Chemical Information Database (IUCLID)  
Japan GHS Classification  
Australia National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
NIOSH (National Institute for Occupational Safety and Health)  
National Library of Medicine's ChemID Plus (NLM CIP)  
National Library of Medicine's PubMed database (NLM PUBMED)  
U.S. National Toxicology Program (NTP)  
New Zealand's Chemical Classification and Information Database (CCID)  
Organization for Economic Co-operation and Development Environment, Health, and Safety Publications  
Organization for Economic Co-operation and Development High Production Volume Chemicals Program  
Organization for Economic Co-operation and Development Screening Information Data Set  
RTECS (Registry of Toxic Effects of Chemical Substances)  
World Health Organization

**Prepared By** Bio-Rad Laboratories, Environmental Health and Safety.**Revision date** 14-Aug-2024**Revision Note** Significant changes throughout SDS. Review all sections.**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

**End of Safety Data Sheet**



# SAFETY DATA SHEET

This safety data sheet was created pursuant to the requirements of:  
Canada WHMIS 2015 which includes the amended Hazardous Products Act (HPA) and the Hazardous Products Regulation (HPR)

Revision date 14-Aug-2024

Revision Number 1.1

## 1. Identification

### Product identifier

Product Name R7b - Conjugate Diluent, 25 ml

### Other means of identification

Catalog Number(s) 7441H

### Recommended use of the chemical and restrictions on use

Recommended use Restricted to professional users  
In vitro diagnostic

Restrictions on use No information available

### Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer Address

Bio-Rad  
3 boulevard Raymond Poincaré  
92430 Marnes-la-Coquette  
France  
e-mail: fds-msds.fr@bio-rad.com

#### Legal Entity / Contact Address

Bio-Rad Laboratories (Canada) Ltd.  
2403 Guenette  
Montreal, Quebec H4R 2E9  
Canada

Technical Service 1-800-361-1808  
CSD\_Techsupport@bio-rad.com

### Emergency telephone number

24 Hour Emergency Phone Number CHEMTREC Canada:1 (800) 424-9300

## 2. Hazard(s) identification

### Classification

|                    |            |
|--------------------|------------|
| Skin sensitization | Category 1 |
|--------------------|------------|

### Label elements

#### **Warning**

#### **Hazard statements**

May cause an allergic skin reaction

**Precautionary Statements - Prevention**

Avoid breathing dust/fume/gas/mist/vapors/spray  
Contaminated work clothing should not be allowed out of the workplace  
Wear protective gloves/protective clothing/eye protection/face protection

**Skin**

IF ON SKIN: Wash with plenty of water and soap  
If skin irritation or rash occurs: Get medical advice/attention  
Take off contaminated clothing and wash it before reuse

**Precautionary Statements - Disposal**

Dispose of contents/container in accordance with local, regional, national, and international regulations as applicable

**Other information**

No information available.

**3. Composition/information on ingredients****Substance**

Not applicable.

**Mixture**

The product contains no substances which at their given concentration, are considered to be hazardous to health

| Chemical name                                                                       | CAS No.    | Weight-%     | Hazardous Material<br>Information Review Act<br>registry number<br>(HMIRA registry #) | Date HMIRA filed and<br>date exemption granted<br>(if applicable) |
|-------------------------------------------------------------------------------------|------------|--------------|---------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with<br>2-methyl-3(2H)-isothiazolone | 55965-84-9 | 0.001 - 0.01 | -                                                                                     |                                                                   |

**4. First-aid measures****Description of first aid measures****General advice**

Show this safety data sheet to the doctor in attendance.

**Inhalation**

Remove to fresh air.

**Eye contact**

Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids.  
Consult a physician.

**Skin contact**

Wash with soap and water. May cause an allergic skin reaction. In the case of skin irritation  
or allergic reactions see a physician.

Ingestion Rinse mouth thoroughly with water.

**Most important symptoms and effects, both acute and delayed**

Symptoms Itching. Rashes. Hives.

**Indication of any immediate medical attention and special treatment needed**

Note to physicians May cause sensitization in susceptible persons. Treat symptomatically.

## 5. Fire-fighting measures

Suitable Extinguishing Media Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

Unsuitable extinguishing media No information available.

Specific hazards arising from the chemical Product is or contains a sensitizer. May cause sensitization by skin contact.

**Explosion data**

Sensitivity to mechanical impact None.

Sensitivity to static discharge None.

Special protective equipment for fire-fighters Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment.

## 6. Accidental release measures

**Personal precautions, protective equipment and emergency procedures**

Personal precautions Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak.

**Methods and material for containment and cleaning up**

Methods for containment Prevent further leakage or spillage if safe to do so.

Methods for cleaning up Pick up and transfer to properly labeled containers.

## 7. Handling and storage

**Precautions for safe handling**

Advice on safe handling Handle in accordance with good industrial hygiene and safety practice. Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. In case of insufficient ventilation, wear suitable respiratory equipment. Do not eat, drink or smoke when using this product. Take off contaminated clothing and wash before reuse.

**Conditions for safe storage, including any incompatibilities**

Storage Conditions Keep containers tightly closed in a dry, cool and well-ventilated place. Store locked up. Keep out of the reach of children. Store according to product and label instructions.

## 8. Exposure controls/personal protection

**Control parameters**

**Exposure Limits** This product, as supplied, does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.

**Appropriate engineering controls**

**Engineering controls** Showers  
Eyewash stations  
Ventilation systems.

**Individual protection measures, such as personal protective equipment**

**Eye/face protection** Wear safety glasses with side shields (or goggles).

**Hand protection** Wear suitable gloves.

**Skin and body protection** Wear suitable protective clothing.

**Respiratory protection** No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

**General hygiene considerations** Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

**Information on basic physical and chemical properties**

**Physical state** Liquid  
**Appearance** Liquid  
**Color** light yellow  
**Odor** Low  
**Odor threshold** No information available

| <b><u>Property</u></b>                         | <b><u>Values</u></b> | <b><u>Remarks • Method</u></b> |
|------------------------------------------------|----------------------|--------------------------------|
| <b>pH</b>                                      | 7.4                  | None known                     |
| <b>Melting point / freezing point</b>          | No data available    | None known                     |
| <b>Initial boiling point and boiling range</b> | No data available    | None known                     |
| <b>Flash point</b>                             | No data available    | None known                     |
| <b>Evaporation rate</b>                        | No data available    | None known                     |
| <b>Flammability</b>                            | No data available    | None known                     |
| <b>Flammability Limit in Air</b>               |                      | None known                     |
| <b>Upper flammability or explosive limits</b>  | No data available    |                                |
| <b>Lower flammability or explosive limits</b>  | No data available    |                                |
| <b>Vapor pressure</b>                          | No data available    | None known                     |
| <b>Relative vapor density</b>                  | No data available    | None known                     |
| <b>Relative density</b>                        | No data available    | None known                     |
| <b>Water solubility</b>                        | Miscible in water    |                                |
| <b>Solubility in other solvents</b>            | No data available    | None known                     |
| <b>Partition coefficient</b>                   | No data available    | None known                     |
| <b>Autoignition temperature</b>                | No data available    | None known                     |
| <b>Decomposition temperature</b>               |                      | None known                     |
| <b>Kinematic viscosity</b>                     | No data available    | None known                     |
| <b>Dynamic viscosity</b>                       | No data available    | None known                     |

**Other information**

**Explosive properties** Not applicable.

|                      |                 |
|----------------------|-----------------|
| Oxidizing properties | Not applicable. |
| Softening point      | Not applicable  |
| Molecular weight     | Not applicable  |
| VOC content          | Not applicable  |

## 10. Stability and reactivity

|                                    |                                           |
|------------------------------------|-------------------------------------------|
| Reactivity                         | No information available.                 |
| Chemical stability                 | Stable under normal conditions.           |
| Possibility of hazardous reactions | None under normal processing.             |
| Conditions to avoid                | None known based on information supplied. |
| Incompatible materials             | None known based on information supplied. |
| Hazardous decomposition products   | None known based on information supplied. |

## 11. Toxicological information

### Information on likely routes of exposure

#### Product Information

|              |                                                                                                                                                                                                                           |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Inhalation   | Specific test data for the substance or mixture is not available.                                                                                                                                                         |
| Eye contact  | Specific test data for the substance or mixture is not available.                                                                                                                                                         |
| Skin contact | May cause sensitization by skin contact. Specific test data for the substance or mixture is not available Repeated or prolonged skin contact may cause allergic reactions with susceptible persons (based on components). |
| Ingestion    | Specific test data for the substance or mixture is not available.                                                                                                                                                         |

### Symptoms related to the physical, chemical and toxicological characteristics

|          |                         |
|----------|-------------------------|
| Symptoms | Itching. Rashes. Hives. |
|----------|-------------------------|

### Acute toxicity

#### Numerical measures of toxicity

No information available

#### Component Information

| Chemical name                                                                                  | Oral LD50          | Dermal LD50              | Inhalation LC50 |
|------------------------------------------------------------------------------------------------|--------------------|--------------------------|-----------------|
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with 2-methyl-3(2H)-isothiazolone<br>55965-84-9 | = 53 mg/kg ( Rat ) | = 87.12 mg/kg ( Rabbit ) | -               |

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                                   |                                                                   |
|-----------------------------------|-------------------------------------------------------------------|
| Skin corrosion/irritation         | Based on available data, the classification criteria are not met. |
| Serious eye damage/eye irritation | Based on available data, the classification criteria are not met. |
| Respiratory or skin sensitization | May cause sensitization by skin contact.                          |
| Germ cell mutagenicity            | Based on available data, the classification criteria are not met. |

|                                 |                                                                   |
|---------------------------------|-------------------------------------------------------------------|
| <b>Carcinogenicity</b>          | Based on available data, the classification criteria are not met. |
| <b>Reproductive toxicity</b>    | Based on available data, the classification criteria are not met. |
| <b>STOT - single exposure</b>   | Based on available data, the classification criteria are not met. |
| <b>STOT - repeated exposure</b> | Based on available data, the classification criteria are not met. |
| <b>Aspiration hazard</b>        | Based on available data, the classification criteria are not met. |

## 12. Ecological information

|                                      |                                                    |
|--------------------------------------|----------------------------------------------------|
| <b>Ecotoxicity</b>                   | Harmful to aquatic life with long lasting effects. |
| <b>Persistence and degradability</b> | No information available.                          |
| <b>Bioaccumulation</b>               | There is no data for this product.                 |

### Component Information

| Chemical name                                                                                     | Partition coefficient |
|---------------------------------------------------------------------------------------------------|-----------------------|
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with<br>2-methyl-3(2H)-isothiazolone<br>55965-84-9 | 0.7                   |

|                              |                           |
|------------------------------|---------------------------|
| <b>Other adverse effects</b> | No information available. |
|------------------------------|---------------------------|

## 13. Disposal considerations

### Disposal methods

|                                            |                                                                                                                 |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>Waste from residues/unused products</b> | Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation. |
| <b>Contaminated packaging</b>              | Do not reuse empty containers.                                                                                  |

## 14. Transport information

|                    |               |
|--------------------|---------------|
| <b><u>TDG</u></b>  | Not regulated |
| <b><u>DOT</u></b>  | Not regulated |
| <b><u>MEX</u></b>  | Not regulated |
| <b><u>IATA</u></b> | Not regulated |
| <b><u>IMDG</u></b> | Not regulated |

## 15. Regulatory information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Regulations

**The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable

The Stockholm Convention on Persistent Organic Pollutants Not applicable

The Rotterdam Convention Not applicable

#### International Inventories

Contact supplier for inventory compliance status

## 16. Other information

|             |                         |                       |                           |                                           |
|-------------|-------------------------|-----------------------|---------------------------|-------------------------------------------|
| <b>NFPA</b> | <b>Health hazards</b> 2 | <b>Flammability</b> 0 | <b>Instability</b> 0      | <b>Physical and chemical properties -</b> |
| <b>HMIS</b> | <b>Health hazards</b> 2 | <b>Flammability</b> 0 | <b>Physical hazards</b> 0 | <b>Personal protection</b> X              |

#### Key or legend to abbreviations and acronyms used in the safety data sheet

##### Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | Sk*  | Skin designation                 |

#### Key literature references and sources for data used to compile the SDS

Agency for Toxic Substances and Disease Registry (ATSDR)  
 U.S. Environmental Protection Agency ChemView Database  
 European Food Safety Authority (EFSA)  
 Environmental Protection Agency  
 Acute Exposure Guideline Level(s) (AEGL(s))  
 U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
 U.S. Environmental Protection Agency High Production Volume Chemicals  
 Food Research Journal  
 Hazardous Substance Database  
 International Uniform Chemical Information Database (IUCLID)  
 Japan GHS Classification  
 Australia National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
 NIOSH (National Institute for Occupational Safety and Health)  
 National Library of Medicine's ChemID Plus (NLM CIP)  
 National Library of Medicine's PubMed database (NLM PUBMED)  
 U.S. National Toxicology Program (NTP)  
 New Zealand's Chemical Classification and Information Database (CCID)  
 Organization for Economic Co-operation and Development Environment, Health, and Safety Publications  
 Organization for Economic Co-operation and Development High Production Volume Chemicals Program  
 Organization for Economic Co-operation and Development Screening Information Data Set  
 RTECS (Registry of Toxic Effects of Chemical Substances)  
 World Health Organization

**Prepared By** Bio-Rad Laboratories, Environmental Health and Safety.

**Revision date** 14-Aug-2024

**Revision Note** Significant changes throughout SDS. Review all sections.

#### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

**End of Safety Data Sheet**